

Hausdorff Dimensions of Quasi-Fuchsian Limit Sets Constructed via Schottky Groups

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Introduction

Fractal geometry is a relatively new field with the word first being used in 1975 to describe rough geometrical shapes (12) (13). The term describes sets with infinite depth and complexity, with origins of these ideas deriving from the 19th century in the theory of non-differentiable functions (5). As the subject has flourished over the past half-century, more interest has been directed towards the field of fractal geometry, either as an interest of mathematical artwork (19) or as an interest of understanding their structure.

To analyse fractals effectively, new applications of mathematical tools have to be introduced such that they are consistent under infinitely detailed systems. With this, a new idea of dimension must be considered, with fractals famously being of fractional dimension (14). It is also important to be able to properly understand a typical fractal construction via a set of generating functions.

This project will work with a set of fractals constructed by Möbius maps, of which the details of Möbius maps are consolidated in the work by Kisil (11), on the Riemann sphere - a one-point compactification of the complex space including the point of infinity. A collection of Möbius maps, used to generate a Schottky group, is chosen such that the interiors of their isometric circles on the Riemann sphere do not intersect. Its limit set when the disks construct a necklace on the space is a fractal homeomorphic to a circle and is called a quasi-circle (16, p. 39)). Bowen has stated the fractal dimension of the limit set $\dim_H(\lambda)$ is known to be bounded by $1 \leq D < 2$ due to the inherent topology of quasi-circles (2, pp. 10-25).

It is the objective of this project to be able to provide rigorous foundations for approaching these limit sets and their Hausdorff dimensions. Included is constructing theoretical knowledge from fields such as fractal analysis, topology, measure theory, and more. Beyond analysis of Möbius map dynamics and the theory of Schottky groups, this project will showcase the several limit sets constructable by Schottky groups and provide an overview of their Hausdorff dimensions.

1 Prerequisite Problems

To be able to approach the project content, it is first of interest to develop an understanding in analysis, geometry, measure theory, and beginner fractal analysis. In this section are the written solutions to some of the problems in the problem sheets provided throughout the early stages of this project, with all work being entirely my own.

Question 1. *Show that the Middle Third Cantor set, constructed by removing open intervals from $[0, 1]$ is a non-empty set whose Lebesgue measure is zero.*



Figure 1: Image showcasing the iterations of the Cantor set from C_0 to C_5 . Image source: (15)

Defining the length of an open interval as $l((a, b)) = b - a$, the Lebesgue measure of a set X is defined to be

$$\lambda(X) = \inf \left\{ \sum_{n=1}^{\infty} l(I_n) : I_n \text{ is a sequence of open intervals such that } X \subset \bigcup_{n=1}^{\infty} I_n \right\}.$$

We may also know that the Cantor set is treated as the limit of the sequence with $C_0 = [0, 1]$:

$$C_n = \frac{C_{n-1}}{3} \cup \frac{C_{n-1} + 2}{3}$$

As a result, we may say that each iteration C_n contains 2^n intervals of length 3^{-n} . We construct a relationship between the Lebesgue measure of the Cantor sequence such that $\lambda(C_n) = \frac{2}{3}C_{n-1}$. When $\lambda(C_0) = 1$ trivially by the definition of the Cantor sequence, we see that the Cantor set has measure

$$\lambda(C) = \lim_{n \rightarrow \infty} (\lambda(C_n)) = \lim_{n \rightarrow \infty} \left(\frac{2}{3} \right)^n = 0.$$

Question 2. Show that the Middle Third Cantor set, constructed by removing open intervals from $[0, 1]$ coincides with the set of those $x \in [0, 1]$ whose base 3 expansion does not include the digit 1.

The middle-third Cantor set is defined by removes the middle third of all current intervals. If we choose to work in base 3 and the Cantor sequence with an initial interval $[0, 1]$, then we see that we may segment the interval into three intervals of equal length: $[0.0_3, 0.1_3] \cup [0.1_3, 0.2_3] \cup [0.2_3, 1.0_3]$. We see that with these equal sized sets, the middle third Cantor sequence should remove the interval $[0.1_3, 0.2_3]$ of all intervals containing the digit 1. It is then required to consider whether the interval $[0.0_3, 0.1_3]$ is permitted, which is the case because $0.1_3 = 0.022 \dots_3$, thus $[0.0_3, 0.1_3]$ does not contain the digit 1.

Question 3. For each $t \in \mathbb{R}$, let's define $C_t := \{x \in C \mid x > t\}$, where C is the Middle Third Cantor set. (a) Give an example of $s \in C$ such that $C_s \subset C$ is a closed set. (b) Give an example of $s \in C$ such that $C_s \subset C$ is not a closed set.

The Cantor set defined by the Cantor sequence iteratively applies scaling and transformation to the initial closed interval $[0, 1]$ with functions $F_1(X) = \frac{X}{3}$ and $F_2(X) = \frac{X+2}{3}$. Since C is the collection of infinite compositions of F_1 and F_2 and these maps both preserve closedness of a set, C is itself closed. The set C_s is guaranteed to be closed so long as we ensure that s is not an end point of an interval I obtained via the constructing sequence of C such that $s \in I = [s, x] : s < x \leq 1$. In the case that s is an end point, then the interval in the constructing sequence will not be closed, therefore the images of that interval and C_s itself will not be closed. An example of s which has C_s closed is the point $s = \frac{1}{3}$. Then, $C_s = C \cap (\frac{1}{3}, 1] = \frac{C+2}{3}$, which is closed because it is the homeomorphism of the closed set C .

Alternatively, an example of s which leaves C_s not closed is the point $s = \frac{2}{3}$. We see that $s = \frac{2}{3}$ is the endpoint of the interval $F_2([0, 1]) = [\frac{2}{3}, 1]$, therefore the constructing sequence does not contain all closed sets and C_s is not closed. We may also prove this value of s breaks the closed property of C_s by noting that s is the limit point of the sequence $F_2 F_1 F_1 F_1 \cdots ([0, 1])$. Closed sets must contain all of the limit points to Cauchy sequences defined within its domain, therefore since $s \notin C_s$, the limit point is not contained in the set and C_s is not closed.

Question 4. Let C be the Middle Third Cantor set. Define a homeomorphism $g : C \times C \rightarrow C$. Is g Lipschitz? In other words, does there exist $k > 0$ such that for all $(x, y) \in C \times C$ we have $|g(x) - g(y)| \leq k||x - y||$.

We observe that since C is compact and totally disconnected in metric space, the product space $C \times C$ must also be compact and disconnected, therefore a homeomorphism must exist. We shall now observe the Hausdorff dimension of the Cantor set $\dim_H(C) = \frac{\ln 2}{\ln 3}$ and the dimension of the product space $\dim_H(C \times C) = 2 \cdot \frac{\ln 2}{\ln 3}$.

We know, however, that if a function g is Lipschitz, then the Hausdorff dimension of its image cannot exceed the Hausdorff dimension of the preimage, and this shall be proven now. The Hausdorff measure is $\mathcal{H}^d(X) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^d(X)$ and, for an open cover $\{B_i\}_{i=1}^\infty$:

$$\mathcal{H}_\delta^d(X) = \inf \left\{ \sum_{i=1}^\infty \text{diam}(B_i)^d : X \subseteq \bigcup_{i=1}^\infty B_i, \text{diam}(B_i) \leq \delta \right\}$$

If g is Lipschitz with constant k , then $\text{diam}(g(B_i)) \leq k \cdot \text{diam}(B_i)$. Now we let $\{B_i\}$ be a δ -cover of X , with $\{g(B_i)\}$ a $k\delta$ -cover of $g(X)$. We then see that $\sum \text{diam}(g(B_i))^d \leq k^d \sum \text{diam}(B_i)^d$ and, after taking the limit of $\delta \rightarrow 0$:

$$\mathcal{H}^d(g(X)) \leq k^d \mathcal{H}^d(X)$$

For any value $d > \dim_H(X)$, then $\mathcal{H}^d(g(X)) = 0$. As a result, the Hausdorff dimension of the image $g(X)$, defined to be the supremum of $\mathcal{H}^d(g(X)) = \infty$, cannot be greater than the dimension of its preimage. Therefore, g is not Lipschitz.

Question 5. Let $0 < \lambda < \eta < 1$. Show that the middle- λ and middle- η Cantor sets are homeomorphic by a Hölder continuous map, and state the Hölder exponent.

We first begin with the definition of what these Cantor sets are. The middle- λ Cantor set, C_λ , is a set constructed by the Cantor sequence $C_{n+1} = F_1(C_n) \cup F_2(C_n)$. The functions F_1, F_2 are composed of scaling and translation such that for an interval $[0, 1]$: $d(F_1([0, 1]), F_2([0, 1])) = \lambda$. We can use this to determine that the scale factor of F_1, F_2 for the middle- λ Cantor set is $S_\lambda = \frac{1-\lambda}{2}$, and the scale factor of F_1, F_2 for the middle- η Cantor set is $S_\eta = \frac{1-\eta}{2}$. Any point in the middle- λ

Cantor set x can be uniquely represented by an infinite sequence $\{a_1, a_2, \dots : a_i \in \{0, 1\}\}$, dictating whether F_1 or F_2 was applied at a specific iteration of the Cantor sequence. We see that

$$x = (1 - S_\lambda) \sum_{k=1}^{\infty} a_k S_\lambda^{k-1}.$$

Then, we choose a function h which maps the middle- λ Cantor set to the middle- η Cantor set such that

$$h(x) = (1 - S_\eta) \sum_{k=1}^{\infty} a_k S_\eta^{k-1}.$$

To determine whether h is Hölder continuous and, if so, what the exponent is, we must first be able to compute the distance between points. Assume that $x, y \in C_\lambda$ are two points which share the same sequence up until the n 'th term. It must then be that the distance between the points is at least the gap between the two images in the n 'th term $|x - y| \geq \lambda S_\lambda^{n-1}$. Similarly, the distance between their images can be no more than the width of the interval in the $n - 1$ 'th interval $|h(x) - h(y)| \leq S_\eta^{n-1}$. To satisfy the Hölder condition, we require an exponent α such that $S_\eta^{n-1} \leq C(\lambda S_\lambda^{n-1})^\alpha$. This can be computed to find that h is a Hölder continuous map with exponent

$$\alpha = \frac{\ln S_\eta}{\ln S_\lambda}.$$

Question 6. *Show that the middle third Cantor set is perfect and nowhere dense.*

We shall begin with showing the Cantor set is perfect by showing it is closed and has no isolated points. It is known that the Cantor set is compact because it is the collection of homeomorphic images of a compact interval. To find whether there are no isolated points, choose two points $x, y \in C$ such that $|x - y| < \epsilon$ for any arbitrarily small epsilon. We shall choose x, y to be represented by an infinite sequence which corresponds to whether the generating function F_1 or F_2 shall be applied, and we choose these sequences for x and y to which they are identical for the first $n + 1$ term. We choose $\epsilon = 3^{-n}$, then it is clear that $|x - y| \leq 3^{-(n+1)} \leq \epsilon$.

To show that the set is nowhere dense, we observe whether C has non-empty interior. We may simply state that the Hausdorff dimension of the Cantor set is below 1. Since the topological dimension is no greater than the Hausdorff dimension, all subsets of C are simply collections of distinct points of no interior.

Question 7. *We can say that a bounded set $X \in \mathbb{R}^d$ has zero d -dimensional area if for any $\epsilon > 0$ we can cover X by a countable family of open balls $\{B_n\}_{n=1}^{\infty}$ whose d -dimensional area is denoted $\mu(B_n)$ such that $\sum_{n=1}^{\infty} \mu(B_n) < \epsilon$. (a) Show that the Cantor set has zero 1-dimensional area. (b) Show the Sierpinski triangle has zero 2-dimensional area.*

To show the Cantor set has zero 1-dimensional area, we simply use the property earlier that the Cantor set is nowhere dense. This implies that there is no interval inside C . We then choose an arbitrary $\epsilon > 0$ and let the collection of the countable points $\{x_i\}_{i=1}^{\infty} \in C$ be covered by a countable set of ball covers $\{B_i\}_{i=1}^{\infty}$. Choose the ball cover such that its 1-dimensional measure is $\text{diam}(B_i) = 2^{-(i+1)} \cdot \epsilon$, then $\sum_{n=1}^{\infty} \text{diam}(B_i) = \frac{\epsilon}{2} < \epsilon$. Therefore, the 1-dimensional measure of the Cantor set is zero.

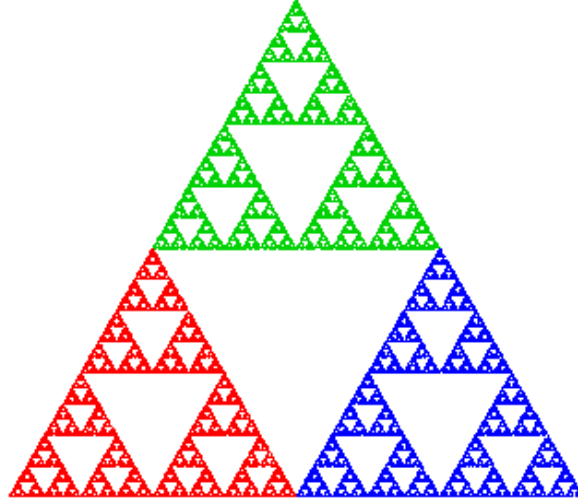


Figure 2: Image of the Sierpinski Triangle, showcasing the images of the set under F_1, F_2, F_3 as different colours. Image source: (20)

The Sierpinski triangle is constructed with an initial equilateral triangle of side length 1, then constructing a sequence which separates the triangle into 3 triangles of side length half of the previous triangle. The Sierpinski triangle is the intersection of all elements of the sequence. Visually, this removes a triangle from the middle of all triangles previously in the set, and can be generated with 3 transformations F_1, F_2, F_3 of scale factor $\frac{1}{2}$. We shall recall the construction sequence of S :

$$S_{n+1} = F_1(S_n) \cup F_2(S_n) \cup F_3(S_n)$$

Observe the difference in 2-dimensional measure between S_n and S_{n+1} :

$$\text{Area}(S_n) = k, \text{Area}(S_{n+1}) = 3 \cdot \frac{k}{2^2} = 0.75k.$$

This sequence of areas also holds for the areas of all covering balls for the sequence element $S_n \subset \{B_i^{(n)}\}_{i=1}^{3^n}$. This is true because for $T_i^{(n)}$ a triangle in S_n and $B_i^{(n)}$ a ball cover of $T_i^{(n)}$, there exists a constant C such that

$$\frac{\text{Area}(B_i^{(n)})}{\text{Area}(T_i^{(n)})} = C.$$

Scaling of the initial area leaves the following iterative sequence invariant; therefore, the sequence of areas also applies to the ball cover. As a result, we may choose arbitrary $\epsilon > 0$ and since $\lim_{n \rightarrow \infty} \sum_{i=1}^{3^n} \text{Area}(B_i^{(n)}) = 0$, there must exist some value of n such that the sum of the areas of balls is less than ϵ . This shows that the 2-dimensional area of the Sierpinski triangle is zero.

For future questions, we shall skip the step for the ball cover and simply observe the sequence of the areas. Since the ball cover follows the same sequence as elements in the sequence, the results of this adjusted method are left unchanged.

Question 8. Consider the Sierpinski carpet. Describe the set corresponding to the intersection with the vertical line of symmetry L , that is $L = \{(x, y) \mid x = \frac{1}{2}\}$

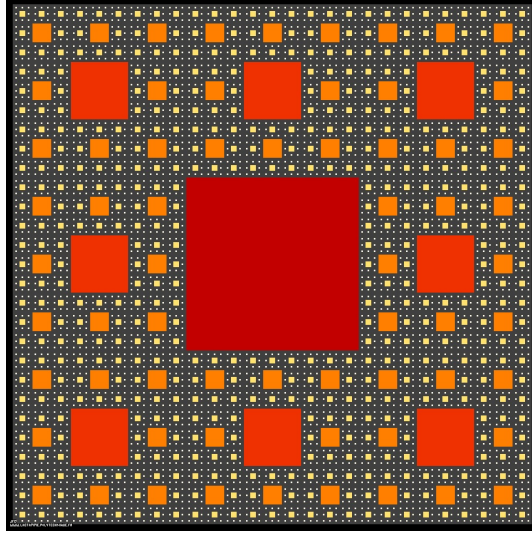


Figure 3: Image of the Sierpinski Carpet at the fifth iteration of its constructing sequence provided in the next question. Image source: (3)

The Sierpinski carpet is similar to the Sierpinski triangle, starting with a square of side length 1 then dividing it into 9 squares of side length $\frac{1}{3}$ and removing the central square. We then see that all points in the Sierpinski carpet can be represented by two infinite sequences $(a_1, a_2, \dots)$ and $(b_1, b_2, \dots)$ such that $a_i, b_i \in \{0, 1, 2\}$. In this case, 0 corresponds to the leftmost column/row, and 2 corresponds to the rightmost column/row, therefore it is by the definition of the Sierpinski carpet that for the n 'th term of the sequences, $(a_n, b_n) \neq (1, 1)$. The line L has non empty interior only when all $a_n = 1$. Then, following the definition of the carpet, $b_n \in \{0, 2\}$. It was proven in Question 2 that the set satisfying this condition of no 1's in the base 3 expansion is the Cantor set. As a result, the intersection of the Sierpinski carpet and the line L is the Cantor set.

Question 9. *Show that the Sierpinski triangle and Sierpinski carpet are not homeomorphic.*

Recall the Sierpinski triangle's constructing sequence via three transformations of disjoint images:

$$S_{n+1} = F_1(S_n) \cup F_2(S_n) \cup F_3(S_b) \subset S_n$$

Using cut point topology, we see that the Sierpinski triangle can be disconnected with the removal of two points - the midpoints of two sides of the initial equilateral triangle in the constructing sequence denoted S_0 . The first iteration of the sequence $S_1 = F_1(S_0) \cup F_2(S_0) \cup F_3(S_0)$ with these points removed has a disjoint triangle, therefore all of S has a disjoint section with these two points removed. This prospect of finite cut points, however, is not possible with the Sierpinski carpet, and requires the removal of an entire interval. Therefore, since they do not share the same cut point topology, they are not homeomorphic.

Question 10. *Let T_n be the set of 3^n smaller triangles used in the construction of the Sierpinski triangle. (a) What is the total perimeter of T_n as $n \rightarrow \infty$? (b) What is the total area of T_n as $n \rightarrow \infty$?*

Every iteration of T_n reduces the size of the triangle sides by a factor $\frac{1}{2}$. It is known that T_0 is the initial equilateral triangle of side lengths 1 and perimeter $P(T_0) = 3$, then $P(T_1) = 3 \cdot (3 \cdot \frac{1}{2}) = 1.5 \cdot P(T_0)$. Generally, the perimeter of T_n is $P(T_n) = 1.5^n \cdot 3$. As $n \rightarrow \infty$, the sequence of perimeters of T_n shall approach ∞ .

Alternatively, the area of a triangle is reduced by a factor $(\frac{1}{2})^2 = \frac{1}{4}$. As a result, given a starting area of $A(T_0) = \frac{\sqrt{3}}{4} = k$, then $A(T_1) = 3 \cdot (\frac{k}{4}) = 0.75 \cdot A(T_0)$. Generally, the area of T_n is $A(T_n) = 0.75^n \cdot \frac{\sqrt{3}}{4}$. As $n \rightarrow \infty$, the sequence of areas of T_n shall approach 0.

Question 11. Let S_n be the set of 8^n smaller squares used in the construction of the Sierpinski carpet. (a) What is the total perimeter of S_n as $n \rightarrow \infty$? (b) What is the total area of S_n as $n \rightarrow \infty$?

The Sierpinski carpet is constructed with a sequence which, in every iteration, maps every square in the preimage to 8 smaller squares with side lengths of scale factor $\frac{1}{3}$. The initial set $S_0 = [0, 1]^2$ has perimeter $P(S_0) = 4$. The sequence requires more attention than an iterative process because under the Sierpinski carpet's constructing sequence, some edges that would be on the perimeter of S_n will be mapped to the interior of S_{n+1} and therefore do not contribute to the perimeter of S_{n+1} . We see that S_1 contains 1 empty square of side length 3^{-1} to consider in the middle, so $P(S_1) = 4 + 4 \cdot 3^{-1}$. Following this, S_2 contains 1 empty square of side length 3^{-1} and 8 empty squares of side length 3^{-2} , thus $P(S_2) = 4 + 4 \cdot 3^{-1} + 8 \cdot 3^{-2}$. S_3 contains 8^2 new empty squares of side length 3^{-3} , and we may now see that the perimeter of a sequence element is

$$P(S_n) = 4 \cdot \frac{4}{3} + \frac{1}{3} \sum_{i=1}^{n-1} \left(\frac{8}{3}\right)^i.$$

Since $\frac{8}{3} > 1$, the limit as $n \rightarrow \infty$, and therefore the perimeter of the Sierpinski carpet, is equal to ∞ .

The area of the squares are reduced by a factor $\frac{1}{9}$, thus $\text{Area}(S_1) = \frac{8}{9} \cdot \text{Area}(S_0)$. Generally, $\text{Area}(S_n) = (\frac{8}{9})^n \cdot \text{Area}(S_0)$ and $\lim_{n \rightarrow \infty} \text{Area}(S_n) = 0$.

Question 12. The Menger sponge is the three dimensional analogue of the Sierpinski carpet. Of the 27 cubes of third the size of the original, we retain 20. Let M_n be the set of 20^n cubes used in the n 'th level of the construction of the Menger curve. (a) What is the area of M_n as $n \rightarrow \infty$? (b) What is the volume of M_n as $n \rightarrow \infty$?

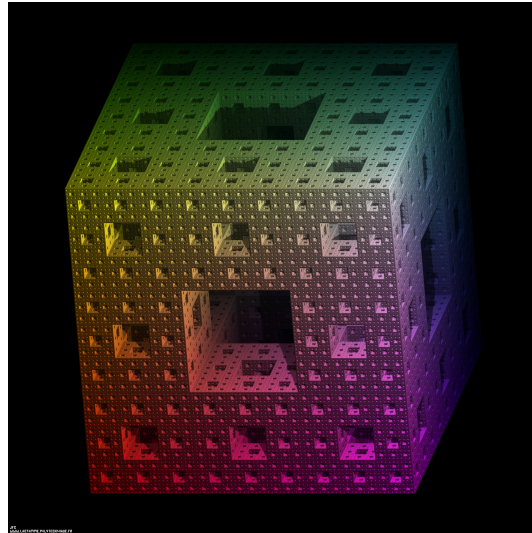


Figure 4: The image of the M_5 using the constructing sequence defined above. Image source: (4)

Let the initial cube M_0 be the unit cube $[0, 1]^3$ such that its surface area is $\text{Area}(M_0) = 6$. The removed smaller cubes in an iteration of the constructing sequence for the Menger cube are the face-centre cubes and the central cube. We may construct a sequence for surface area similar to in the previous question, though it is sufficient to state the known value for the surface area of the n 'th iteration of the Menger cube's constructing sequence which is

$$\text{Area}(M_n) = 2 \cdot \left(\frac{20}{9}\right)^n + 4 \cdot \left(\frac{8}{9}\right)^n.$$

Since $\frac{20}{9} > 1$, the limit as $n \rightarrow \infty$, and therefore the surface area of the Menger cube, is equal to ∞ .

The volume of M_n decreases, however. Applying the constructing sequence, we see that $V(M_0) = 1$, $V(M_1) = \frac{20}{27} \cdot V(M_0)$. Generally, $V(M_n) = \left(\frac{20}{27}\right)^n$ and $\lim_{n \rightarrow \infty} V(M_n) = 0$.

Question 13. Find a sequence of finite subsets of the plane $\mathbb{R}^2$ which converges to the unit square $[0, 1]$ in the Hausdorff metric.

We require a sequence which contains only points, yet converges to the unit square. For this, choose the sequence

$$K_n = \left\{ \left(\frac{i}{n}, \frac{j}{n} \right) : i, j \in \{0, 1, \dots, n\} \right\}.$$

Choosing $\epsilon = \frac{1}{n} > 0$ an arbitrarily small number, then we may see that the n 'th term of the sequence has that any two points next to each other vertically, horizontally or diagonally are no further apart than $\frac{\sqrt{2}}{n}$. As a result, any point in the unit square is no more than $\frac{\sqrt{2}}{2n} < \epsilon$ away from a point in K_n . Since the Hausdorff metric is defined to be the supremum of the distance metric between points in two different sets, $h(K_n, [0, 1]^2) = \frac{\sqrt{2}}{2n} < \epsilon$, and $\lim_{n \rightarrow \infty} K_n = [0, 1]^2$.

Question 14. (a) What is the Hausdorff distance between the circle and one of its diameters? (b) What is the Hausdorff distance between the unit square $[0, 1]^2$ and the unit circle $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$?

The Hausdorff distance is defined to be

$$h(X, Y) = \max \left\{ \sup_{x \in X} \inf_{y \in Y} d(x, y), \sup_{y \in Y} \inf_{x \in X} d(x, y) \right\}.$$

It is trivial to see that for the circle and one of its diameters, the distance between a point and the other set is at its largest when we take the centre of the circle and the point on the circumference which intersects the line perpendicular to the diameter which passes through the centre. This is true for both terms in the Hausdorff distance. The distance is then simply the radius, which is $h(\text{diameter}, \mathbb{S}) = 1$.

For the unit circle and the unit square, consider both terms in the Hausdorff distance. Considering the supremum of a point on the unit circle, we see that since the unit circle is contained within the unit square, there is no point on the circle such that the infimum of a point on the unit square shall be greater than zero. From the other term in the distance, we see that the distance between points is at its supremum when we choose one of the corners of the unit square. If we construct a line from the centre of the circle to our chosen corner, then the nearest point to the corner in the unit circle, and therefore the point we calculate the distance with, is the point that is intersected by this line. This is calculated to be $h([0, 1]^2, \mathbb{S}) = \sqrt{2} - 1$.

Question 15. Let $X = [0, 1]^2$, $Y = \{(0, 0)\}$ and $Z = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. Compute the Hausdorff distances $h(X, Y)$, $h(X, Z)$ and $h(Y, Z)$.

Recall the definition of Hausdorff distance:

$$h(A, B) = \max\left\{\sup_{a \in A} \inf_{b \in B} d(a, b), \sup_{b \in B} \inf_{a \in A} d(a, b)\right\}$$

Regarding $h(X, Y)$, we see that if we start with the second term $\sup_{y \in Y} \inf_{x \in X} d(x, y)$, then $Y \subset X$ and we choose $y = x$, so $\sup_{y \in Y} \inf_{x \in X} d(x, y) = 0$. Considering the first term $\sup_{x \in X} \inf_{y \in Y} d(x, y)$, we choose $x = (1, 1)$, then $d((1, 1), (0, 0)) = h(X, Y) = \sqrt{2}$. For $h(X, Z)$ and its first term $\sup_{x \in X} \inf_{z \in Z} d(x, z)$, we choose $x = (0.5, 0.5)$ and z can be any corner, resulting in $\sup_{x \in X} \inf_{z \in Z} d(x, z) = \frac{1}{\sqrt{2}}$. For the second term, $\sup_{z \in Z} \inf_{x \in X} d(x, z)$, we observe that $Z \subset X$, thus no point can have non-zero distance. Therefore, $h(X, Z) = \frac{1}{\sqrt{2}}$.

The Hausdorff metric between Y and Z , $h(Y, Z)$, has a first term $\sup_{y \in Y} \inf_{z \in Z} d(y, z)$ of 0 because $Y \subset Z$. For the second term, we choose $z = (1, 1)$, resulting in $\sup_{z \in Z} \inf_{y \in Y} d(y, z) = h(Y, Z) = \sqrt{2}$.

Question 16. Show that the countable set $\mathbb{Q} \cap [0, 1]$ has zero 1-dimensional measure. Is this true for any countable set in $[0, 1]$?

The rationals in the domain $[0, 1]$ are countable, which is to say that for $r_i \in \mathbb{Q} \cap [0, 1]$, there exists a countable and disjoint collection of subsets such that $\mathbb{Q} \cap [0, 1] = \bigcup_{i=1}^{\infty} \{r_i\}$. It is then a known property that the measure of a set is equal to the sum of measures of its countable collection of disjoint subsets. The 1-dimensional measure of a point is zero, therefore that is also the case for the rationals within the range $[0, 1]$.

Question 17. Let E denote the set of reals in $[0, 1]$ whose decimal expansion contains only the digits 2, 7, 8. That is,

$$E = \left\{ \sum_{n=1}^{\infty} a_n 10^{-n} : a_n \in \{2, 7, 8\} \right\}.$$

Find $\text{diam}(E) = \sup\{|x - y| : x, y \in E\}$ and show that E is self similar.

It is trivial to see that the largest distance between points is between the point defined by the sequence of all $a_i = 2$ and the point defined by the sequence of all $a_i = 7$. Thus, the diameter is

$$\text{diam}(E) = |0.\bar{7} - 0.\bar{2}| = 0.\bar{5} = \frac{5}{9}.$$

To show that the set is self similar, we shall define a constructing sequence of E such that given $E_0 = 0$:

$$E_{n+1} = (E_n + 2 \cdot 10^{-n}) \cup (E_n + 7 \cdot 10^{-n}) \cup (E_n + 8 \cdot 10^{-n}) =$$

It is then clear that E may be defined to be the limit of this constructing sequence as n approaches infinity. The sequence is bounded within the compact space $[0, 1]$, therefore the sequence must converge to a fixed point E such that if $E_n = E$, then $E_{n+1} = E_n$. This is the definition of self similarity, therefore E is self similar.

Question 18. Give an example of non-empty sets A, B and a space $\mathcal{X}$ such that

$$A \not\subset \{y \in \mathcal{X} : d(A, B) > d(B, y)\}.$$

Choose the space $\mathcal{X} = \mathbb{R}$ and the set $B = [0, 1]$. Then, we may consider the set

$$A = \left\{ a : d(a, B) \geq \frac{1}{2} \right\}.$$

The set of all points y which satisfy the condition $d(A, B) > d(B, y)$ will be the set

$$Y = \left\{ y : d(y, B) < \frac{1}{2} \right\}$$

It is trivial to see that Y is the complement of A and therefore that $A \cap Y = \emptyset$, providing the above condition.

Question 19. Let A be a non-empty subset of $\mathbb{R}^n$. Show that for every $\epsilon > 0$ and $\delta > 0$, we have that $B(B(A, \epsilon), \delta) \subset B(A, \epsilon + \delta)$. Furthermore, give an example of a metric space, a compact non-empty subset A and two numbers $\epsilon, \delta > 0$ such that $B(B(A, \epsilon), \delta) \neq B(A, \epsilon + \delta)$.

Observe the left side of the equality. When in the space $\mathbb{R}^n$, an ϵ -cover of a set A is defined to be

$$B(A, \epsilon) = \left\{ x \in (R)^n : \inf_{a \in A} d(a, x) < \epsilon \right\}.$$

We may use this property to state that $\sup_{a \in A} d(x, a) = \epsilon$. We do the same for the δ -cover of this cover such that $\inf_{a \in B(A, \epsilon)} d(x, a) < \delta$ and $\sup_{a \in B(A, \epsilon)} d(x, a) = \delta$. To find the distance between the new δ -cover and the original set A , we shall use the triangle inequality:

$$d(A, B(B(A, \epsilon), \delta)) \leq d(A, B(A, \epsilon)) + d(B(A, \epsilon), B(B(A, \epsilon), \delta)) < \epsilon + \delta = \sup\{d(A, B(A, \epsilon + \delta))\}$$

As a result, we see that $B(B(A, \epsilon), \delta) \subset B(A, \epsilon + \delta)$.

To find a set and space where the two covers are not equivalent, choose a discrete topology such as $\mathbb{Z}$ and the values $\epsilon, \delta = \frac{2}{3}$. We see that if $A = \{0\}$, then $B(A, \epsilon)$ involves finding all integers within a distance $\frac{2}{3}$ from A . Since there are no integers contained in this range, $B(A, \epsilon) = A$ and, using the same argument for the δ -cover, $B(B(A, \epsilon), \delta) = A$. On the other hand, we see that $\epsilon + \delta = \frac{4}{3}$ and $B(A, \epsilon + \delta) = B(\{0\}, \frac{4}{3}) = \{-1, 0, 1\} \neq B(B(A, \epsilon), \delta)$. Therefore, we have successfully constructed a counterexample to the strict equality.

Question 20. Show that for compact subsets $A, B \subset \mathbb{R}^n$ we have

$$\inf\{\delta > 0 \mid (\forall b \in B \exists a \in A : d(a, b) < \delta) \text{ and } (\forall a \in A \exists b \in B : d(a, b) < \delta)\} = \max\{\sup_{b \in B} \inf_{a \in A} d(a, b), \sup_{a \in A} \inf_{b \in B} d(a, b)\}$$

We shall let the left side of the equality's collection of δ be L and the right side of the equality be R for shorthand, thus it is the goal to show that $\inf L = R$. We first observe that the distance between a point and a set is $d(x, Y) = \inf_{y \in Y} d(x, y)$. If we apply this to A and B , we see that since the sets are compact, the infimum is attainable. The first condition in L states $\forall b \in B, \exists a \in A : d(a, b) < \delta$. This is equivalent to saying that for any $b \in B$, the minimum distance to A is bounded above by δ . For this to hold true,

$$\sup_{b \in B} d(A, b) = \sup_{b \in B} \inf_{a \in A} d(a, b) < \delta$$

. This same argument is applicable to show that the second condition in L is equivalent to saying $\sup_{a \in A} \inf_{b \in B} d(a, b) < \delta$. δ is a constant in L if it satisfies both conditions simultaneously, or that $\max\{\sup_{b \in B} \inf_{a \in A} d(a, b), \sup_{a \in A} \inf_{b \in B} d(a, b)\} < \delta \in L$. We see that $\forall \delta \in L : \delta > R$. We know that both L and R are non negative because they both contain distance functions. Therefore, it is using the property of infinums that $\inf L = R$, proving the required equality.

Question 21. *show that $D(A, B) = \inf_{a \in A} \inf_{b \in B} d(a, b)$ does not satisfy the triangle inequality.*

Choose the space $\mathbb{R}$ and three sets $X = [0, 1]$, $Y = [1, 2]$ and $Z = [2, 3]$. We can compute the distances between all of these sets using the notion of metric D and choosing the two points which minimise the resulting output. We see that $D(X, Y) = d(1, 1) = 0$, $D(Y, Z) = d(2, 2) = 0$, and $D(X, Z) = d(1, 2) = 1$. The triangle inequality states that $D(X, Z) \leq D(X, Y) + D(Y, Z)$, however this cannot be the case because then $1 \leq 0$. Therefore, there is a contradiction and the function $D(A, B)$ does not satisfy the triangle inequality.

Question 22. *Let X, Y be non-empty compact subsets of $\mathbb{R}^n$. Show that $X \subset \overline{B(Y, h(X, Y))}$ (the closure of $B(Y, h(X, Y))$).*

We shall first observe some properties of the closure $\overline{B(Y, h(X, Y))}$, where $h(X, Y)$ is the Hausdorff metric. If $X \subset Y$, then $h(X, Y) = 0$ and $\overline{B(Y, h(X, Y))} = Y \supset X$.

Now consider $X \not\subset Y$. The Hausdorff metric is defined such that if $h(X, Y) = \delta$, then for all points $x \in X$, we have that $d(x, Y) \leq \delta$. Similarly, for all points $y \in Y$, we have $d(X, y) \leq \delta$. Therefore, it is trivial to see that $X \subset \overline{B(Y, \delta)}$, where the closure is required to ensure the furthest point $x_0 : \sup_{x \in X} \inf_{y \in Y} d(x, y) = \inf_{y \in Y} d(x_0, y)$ is contained in the cover. Since $\delta = h(X, Y)$, our case is proven.

Question 23. *Is the box dimension of the unit circle well defined? If so, what is the box dimension?*

The box dimension relies on constructing a cover of boxes with side length $\epsilon > 0$, considering the ratio $\frac{\ln N}{\ln 1/\epsilon}$, where N is the number of boxes needed to cover the set, and finding the limit as $\epsilon \rightarrow 0$. To check if the box dimension is well defined, we must calculate the upper and lower box dimensions which find the supremum and infimum respectively of the limit of ϵ . If the upper and lower box dimension are equal, then the box dimension is well defined as a result of the squeeze theorem and is equal to the upper and lower box dimension.

To find an upper bound for the box dimension, we may want to study a small section of the circle. We choose the upper right quadrant of the unit circle which is written as a function $f(x) = \sqrt{1 - x^2}$ for $x \in [0, 1]$. This section is identical to all others, so it is suitable to work with this. We may find that the arc length of this segment is $\frac{\pi}{2}$. Choosing $\epsilon > 0$ and a collection of boxes of side length ϵ , we see that we require $\frac{\pi}{2\epsilon}$ boxes to cover the subsection, therefore needing $\frac{2\pi}{\epsilon}$ to cover the entire circle. Then, we may calculate the lower box dimension:

$$\underline{\dim}_B(\mathbb{S}) = \lim_{\epsilon \rightarrow 0} \frac{\ln 2\pi + \ln 1/\epsilon}{\ln 1/\epsilon} = 1$$

We may find the upper box dimension by constructing a generous cover of the set. Allow us to construct an annulus which is an ϵ -cover of the unit circle. Then we can calculate the area of the annulus A to be $\text{Area}(A) = \pi[(1 + \epsilon)^2 - (1 - \epsilon)^2] = 4\pi\epsilon$. This needs at least $\frac{4\pi\epsilon}{\epsilon^2} = \frac{4\pi}{\epsilon}$ boxes to cover it, resulting in the fact that

$$\overline{\dim}_B(\mathbb{S}) = \lim_{\epsilon \rightarrow 0} \frac{\ln 4\pi + \ln 1/\epsilon}{\ln 1/\epsilon} = 1$$

From this, we have shown that $\underline{\dim}_B(\mathbb{S}) = \overline{\dim}_B(\mathbb{S})$, therefore the box dimension is well defined and is equal to 1.

Question 24. *Is the box dimension of the Sierpinski triangle well defined? If so, what is the box dimension?*

The Sierpinski triangle is constructed by the sequence $S_{n+1} = F_1(S_n) \cup F_2(S_n) \cup F_3(S_n)$, where F_i are all translation and scaling by a factor $\frac{1}{3}$ such that all images are disjoint and $S_{n+1} \subset S_n$. This means that on the n 'th iteration, there are 3^n triangles of side length 2^{-n} . The Sierpinski triangle is defined as the intersection of all elements of this sequence, or it can be defined as the limit of this sequence as n approaches infinity.

For the lower box dimension, let $\epsilon = 2^{-n} > 0$. Then, for the n 'th element of the sequence, we simply require at most 3^n boxes to cover the set, resulting in a lower box dimension of $\underline{\dim}_B(S) = \lim_{n \rightarrow \infty} \frac{\ln 3^n}{\ln 2^n} = \frac{\ln 3}{\ln 2}$.

For the upper box dimension, we let $2^{-(n+1)} < \epsilon < 2^{-n}$ and we seek to determine the number of boxes required to cover the set. The area of a triangle is larger than the area of the ϵ -box by a factor $C > 1$. Therefore, it must be the case that is at least $C \cdot 3^n$ boxes needed to cover the set. For the denominator, since we wish to find the largest value of the box dimension, it is largest when we choose $\epsilon = 2^{-n}$. This gives the upper box dimension to be:

$$\overline{\dim}_B(S) = \lim_{n \rightarrow \infty} \frac{\ln C + \ln 3^n}{\ln 2^n} = \frac{\ln 3}{\ln 2}$$

Therefore, the box dimension exists and is equal to $\frac{\ln 3}{\ln 2}$

Question 25. *Is the box dimension of the Sierpinski carpet well defined? If so, what is the box dimension?*

Recall that the Sierpinski carpet M iteratively maps a square into 8 disjoint squares in the domain of the preimage with sides of scale factor $\frac{1}{3}$. This can be written as, with $M_0 = [0, 1]^2$:

$$M_{n+1} = F_1(M_n) \cup F_2(M_n) \cup \dots \cup F_8(M_n).$$

For the n 'th element of this sequence, there are 8^n squares of side length 3^{-n} .

To find the lower box dimension, we shall let $\epsilon = 3^{-n}$, then there must be at most 8^n boxes required to cover the Sierpinski carpet. We shall then compute the lower box dimension:

$$\underline{\dim}_B(M) = \lim_{n \rightarrow \infty} \frac{\ln 8^n}{\ln 3^n} = \frac{\ln 8}{\ln 3}$$

For the upper box dimension, assume $2^{-(n+1)} < \epsilon < 2^{-n}$, such that the area of a single triangle has a larger area than an ϵ -box by a scale factor C . Then, there requires at least $C \cdot 8^n$ boxes to cover the Sierpinski carpet. For the denominator, the upper box dimension is largest when we choose to use $\epsilon = 2^{-n}$, then we may compute the upper box dimension: $\overline{\dim}_B(M) = \lim_{n \rightarrow \infty} \frac{\ln C + \ln 8^n}{\ln 3^n} = \frac{\ln 8}{\ln 3}$.

Since the upper and lower box dimension are equal, the box dimension is well defined and is $\dim_B(M) = \frac{\ln 8}{\ln 3}$.

Question 26. *Is the box dimension of the set $K \subset [0, 1]$ whose decimal expansion contains only the numbers $\{0, 2, 5\}$ well defined? If so, what is the box dimension?*

The set K may be defined by a constructing sequence such that $K_0 = [0, 1]$ and $K_{n+1} = \bigcup_{i=1}^3 \frac{K_n + a_i}{10}$, where $a_i \in \{0, 2, 5\}$. We see that K is then the fixed point of this set which may be defined as the limit of the set as $n \rightarrow \infty$. Each interval in K_n has a size 10^{-n} , and K_n has 3^n intervals

Starting with the lower box dimension, we shall let $\epsilon = 10^{-n}$ and see that K_n requires at most 3^n boxes to cover it. As a result, we may then compute the lower box dimension to be:

$$\underline{\dim}_B(K) = \underline{\lim}_{n \rightarrow \infty} \frac{\ln 3^n}{\ln 10^n} = \frac{\ln 3}{\ln 10}$$

For the upper box dimension, we shall let $10^{-(n+1)} < \epsilon < 10^{-n}$ and we see that the intervals in K_n are larger than the boxes by a scale factor C , meaning that there requires at least $C \cdot 3^n$ boxes to cover the set. When computing the upper box dimension, we find the value is maximised if we let $\epsilon = 10^{-n}$ for the denominator. As a result, the upper box dimension is computed to be:

$$\overline{\dim}_B(K) = \overline{\lim}_{n \rightarrow \infty} \frac{\ln C + \ln 3^n}{\ln 10^n} = \frac{\ln 3}{\ln 10}$$

Since the lower and upper box dimensions are equal, the box dimension is well defined and is equal to $\frac{\ln 3}{\ln 10}$.

Question 27. *Show that for any non-trivial compact self-similar set X , the lower box dimension of X is strictly positive such that $\underline{\dim}_B(X) > 0$.*

We shall use the methods of computing the lower box dimension as seen before. A self-similar set implies that it is the fixed point of a constructing sequence. Assuming X is any compact set within the domain of X , we may define this sequence to be $X_{n+1} = \bigcup_{i=1}^r F_i(X_n)$, where the finite collection of r functions F_i are strictly contracting, such as to remain the bounded property of compact sets, and are unique such that $F_i(X_n) \cup F_j(X_n)$ cannot be simplified into its own map $F_x(X_n)$. We can then say that the functions F_i are composed of a scaling of factor $0 < a_i < 1$ to the sides of the sets in X_n .

Assume that X is of topological dimension $k \in \mathbb{N}$ and μ is an k -dimensional measure, then the size of the new iteration scales by a factor $\mu(X_{n+1}) = (\sum_{i=1}^r a_i^k) \cdot \mu(X_n)$. We may use this to calculate the lower box dimension of the set. We let $\epsilon = \max\{a_1, a_2, \dots, a_r\}^n = a^n$, then there requires at most r^n boxes. The lower box dimension can be calculated to be:

$$\underline{\dim}_B(X) = \underline{\lim}_{n \rightarrow \infty} \frac{\ln r^n}{\ln a^n} = \frac{\ln r}{\ln a} > 0$$

Question 28. *Let $n, m > 1$ be integers and let X be the associated Bedford-McMullen carpet. What is the resulting X if we choose only one rectangle, or when we choose every rectangle?*

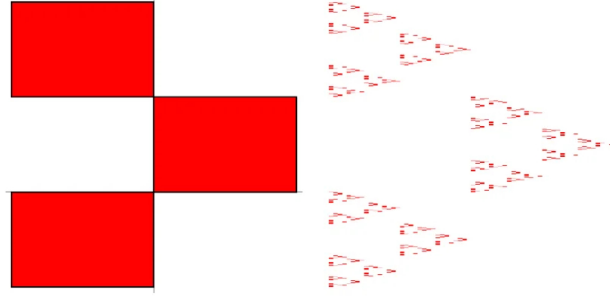


Figure 5: An image of the Bedford-McMullen carpet at $n = 2, m = 3$ and $D = \{(0, 0), (1, 1), (0, 2)\}$. Left side is the first iteration of the generating sequence, and right side is the limit set. Image source: (10)

The Bedford-McMullen carpet is a fractal defined by constructing an $n \times m$ grid on the unit square $[0, 1]^2$ and constructing maps which map the square into rectangles of size $\frac{1}{n} \times \frac{1}{m}$ at a given point (i, j) on the intersecting grid lines which defines where the bottom left corner of the rectangle lies. This may be defined mathematically as saying for a digit set $D = \{(i_n, j_n)\}$, functions $\phi_{ij}(x, y) = (\frac{x+i}{n}, \frac{y+j}{m})$ and the initial set $X_0 = [0, 1]^2$, the fractal is the limit set of the sequence:

$$K_{n+1} = \bigcup_{(i,j) \in D} \phi_{ij}(K_n)$$

If we choose $n, m > 1$ and only one rectangle, this corresponds to only one point (i, j) in D with scale factor $\frac{1}{nm}$ to the area of the set. The resulting sequence means that in X_k , there is only 1 rectangle of size $(\frac{1}{nm})^k$. The limit set is therefore only 1 point, having dimension 0.

If we choose every rectangle, such that for an $n \times m$ grid, D is every possible collection of points (i, j) , then the set K_{n+1} is simply the same as K_n . This can be seen because there is no space in which a rectangle is not stated to be at. As a result, the set is unchanging and the unit square $[0, 1]^2$ is the limit set of dimension 2.

Question 29. For a Bedford-McMullen carpet, choose $n = m = 3$ and choose all squares except the middle one. What is the resulting fractal?

We choose $D = \{(i, j)\}_{i,j=0}^2 \setminus \{(1, 1)\}$, which aligns with this case. Knowing $X_0 = [0, 1]^2$, observing X_1 shows that we map the unit square into 8 squares of scale factor $\frac{1}{3}$ and remove the middle square. This is equivalent to the definition of the Sierpinski carpet, in theory and in the defining sequence, therefore $X \equiv S$.

Question 30. What is the generalised formula for finding the box dimension of a Bedford-McMullen carpet?

We shall let $n, m > 1$ and let $D = \{(i, j)\}$ such that $|D| = k$. Upon every iteration of the constructing sequence $X_{n+1} = \bigcup_{(i,j) \in D} \phi_{ij}(X_n)$, we shall have k rectangles of size $\frac{1}{nm}$. We may use this information to compute the box dimension.

For the lower box dimension, we let $\epsilon = \max\{\frac{1}{n}, \frac{1}{m}\}^n = M^n$, thus we shall require at most k^n boxes. The lower box dimension may be computed to be:

$$\underline{\dim}_B(X) = \lim_{n \rightarrow \infty} \frac{\ln k^n}{\ln M^n} = \frac{\ln k}{\ln M}$$

For the upper box dimension, let $\min\{\frac{1}{n}, \frac{1}{m}\}^n \leq \epsilon < \max\{\frac{1}{n}, \frac{1}{m}\}^n$. If the area of a rectangle is now larger than the area of an ϵ -box by a factor C , then we should require at least $C \cdot k^n$ boxes to cover X . To maximise the upper box dimension, we shall choose $\epsilon = M^n$. As a result, the upper box dimension is:

$$\overline{\dim}_B(X) = \lim_{n \rightarrow \infty} \frac{\ln C + \ln k^n}{\ln M^n} = \frac{\ln k}{\ln M}$$

The two values for the lower and upper box dimension are equal, therefore the box dimension exists and is equal to the ratio of logs for the amount of terms in the digit set D and the largest side of the grid rectangles.

Question 31. Let $0 < \lambda < 1$ and C_λ be the middle- λ Cantor set. Show that the box dimension of C_λ is well-defined, and state its value.

The Cantor set is constructed by two function F_1, F_2 , composed of translation and scaling by a scale factor $\frac{1-\lambda}{2}$, in the sequence

$$C_{\lambda, n+1} = F_1(C_{\lambda, n}) \cup F_2(C_{\lambda, n}).$$

We may use this to calculate the box dimension.

For the lower box dimension, choose $\epsilon = (\frac{1-\lambda}{2})^n$, then there must be at most 2^n boxes required to cover the set:

$$\underline{\dim}_B(C_\lambda) = \lim_{n \rightarrow \infty} \frac{\ln 2^n}{\ln (\frac{2}{1-\lambda})^n} = \frac{\ln 2}{\ln \frac{2}{1-\lambda}}$$

For the upper box dimension, choose $(\frac{1-\lambda}{2})^{n+1} < \epsilon < (\frac{1-\lambda}{2})^n$, then the intervals in $C_{\lambda, n}$ are larger than the ϵ -box by a factor c , therefore requiring at least $C \cdot 2^n$ boxes to cover C_λ :

$$\overline{\dim}_B(C_\lambda) = \lim_{n \rightarrow \infty} \frac{\ln c + \ln 2^n}{\ln (\frac{2}{1-\lambda})^n} = \frac{\ln 2}{\ln \frac{2}{1-\lambda}}$$

Therefore, the box dimension is well defined and is equal to the value above.

It is worth noting that for compact self-similar sets such as the Cantor set, the Hausdorff dimension and box dimension always exist and are equal. As a result, we can also use this formula to calculate the Hausdorff dimension of any Cantor set.

Question 32. Let $X \subset [0, 1]$ with $\overline{\dim}_B(X) < 0.5$ and let $X + \theta X = \{k_1 + \theta k_2 : k_1, k_2 \in X\}$. Show that $X + \theta X$ cannot contain a non-trivial interval.

We shall observe that with the upper box dimension being strictly less than 0.5, then twice its dimension is strictly less than 1. The map $f_\theta : X \times X \rightarrow \mathbb{R}$ defined to be $f_\theta(X, X) = X + \theta X$ is composed of addition and scaling, therefore f_θ is a lipschitz function and

$$\overline{\dim}_B(f_\theta(X)) \leq \overline{\dim}_B(X \times X) = 2 \cdot \overline{\dim}_B(X) < 1.$$

We find that even if X and θX are perfectly disjoint, the dimension will always be less than 1. It is known that the dimension of a set is equal to the supremum of the dimension of its components, therefore it cannot have an interval because intervals have dimension $\dim_B(I) = 1 > 2 \cdot \overline{\dim}_B(X)$.

Question 33. Give an example of a set with Hausdorff dimension of at least 1, yet with zero 1-dimensional measure.

We know that the Hausdorff dimension of Cantor sets is always well-defined and that the dimension of the cartesian product of sets is equal to the sum of dimensions of the individual sets. As a result, we may choose our set to be the cartesian product of Cantor sets $X = C_\lambda \times C_\lambda$ such that $0.5 < \dim_H(C_\lambda) < 1$. When this is the case, then $\dim_H(X) = 2 \cdot \dim_H(C_\lambda) > 1$. However, the measure of the Cantor set is known to be zero and it is trivial to see that the measure of the cartesian product of sets with zero measure must also be of zero measure. Therefore, $X = C_\lambda \times C_\lambda$ is a suitable example for this problem.

Toy Model

To showcase how certain properties of fractals are observed and to give overview of known results, it is important to have a toy model which can provide a simple structure to work with. For this, we will use the Cantor set:

Definition 1. Let $C_0 = [0, 1] \in \mathbb{R}$. The Cantor sequence is defined iteratively as

$$C_{n+1} = \frac{C_n}{3} \cup \frac{2 + C_n}{3}$$

From this definition, we may wish to rewrite the images of C_n as images of functions F_1, F_2 .

Proposition 1. For all elements of the Cantor sequence C_n with two generating functions $F_1(X) = \frac{X}{3}$ and $F_2(X) = \frac{2}{3} + \frac{X}{3}$:

$$C_{n+1} = F_1(C_n) \cup F_2(C_n).$$

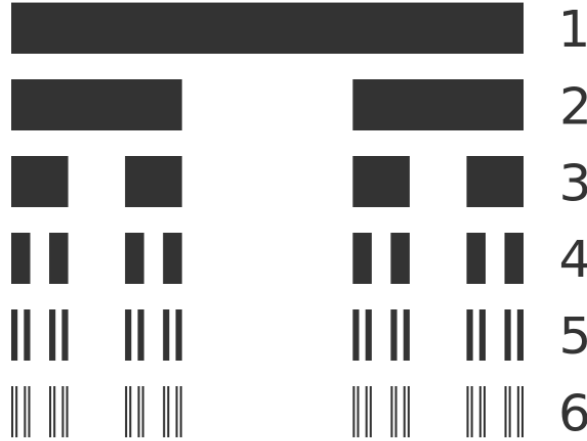


Figure 6: Image of the Cantor sequence up to C_5 . Image source: (15)

Below are the first two terms of the Cantor sequence.

$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$$

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{3}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]$$

Lemma 1. *The sequence of sets $\{C_n\}_{n=0}^\infty$ is nested. For all $n \in \mathbb{N}$, we have that*

$$C_{n+1} \subset C_n.$$

With this lemma, it follows via induction that for all natural numbers n, m such that $n > m$, $C_n \subset C_m$.

Definition 2. *The Cantor set is defined as the intersection of all elements of the Cantor sequence.*

$$C = \bigcap_{n=0}^{\infty} C_n \in \mathbb{R}$$

The Cantor set is easy to analyse as a result of the iterations applying simple functions F_1, F_2 composed of translation and scaling. We will use the definitions for the Cantor sequence and Cantor set to derive further properties.

Theorem 1. *All intervals in an element of the Cantor sequence are disjoint.*

Proof. Prove by induction. Start with the base case of $n = 0$ which is trivially disjoint:

$$C_0 = [0, 1]$$

Then, assume that the condition holds for any C_k such that the intervals in C_k are disjoint. Now, analyse C_{k+1} , constructed via iteration:

$$C_{k+1} = F_1(C_k) \cup F_2(C_k)$$

It is a result of Lemma 1 that the newly constructed intervals in C_{k+1} are contained within C_k , which is already disjoint. Consider a singular closed interval $[a, b] \in C_k$. Applying F_1 and F_2 to $[a, b]$ gives two new intervals:

$$F_1([a, b]) = \left[a, \frac{2a+b}{3} \right], \quad F_2(I) = \left[\frac{a+2b}{3}, b \right]$$

It is trivial that the images of the interval $F_1([a, b]), F_2([a, b])$ are subsets of the original interval $[a, b]$. It also follows from the definition of a, b that $2a + b < a + 2b$ and:

$$F_1([a, b]) \cap F_2([a, b]) = \emptyset.$$

Therefore, the images of an interval under F_1, F_2 are subsets of the interval disjoint to each other, and are thus disjoint from all other intervals in C_{k+1} . □

It is a result of Theorem 1 and Lemma 1 that the Cantor set satisfies the Open Set Condition, requiring that the sequence is nested and has disjointed elements.

Lemma 2. *For the n 'th term of the Cantor sequence, there exists 2^n disjoint intervals of size 3^{-n} .*

Proof. This is a direct application of the generating functions F_1, F_2 . C_0 contains $1 = 2^0$ interval of size $1 = 3^{-0}$. The next iteration of the sequence constructs two disjoint image intervals under F_1, F_2 from each interval in the sequence's prior element. Defining $\#(X)$ as the number of intervals in the set, it is observed that $\#(C_{n+1}) = 2 \cdot \#(C_n)$. Therefore,

$$\#(C_n) = 2^n$$

Furthermore, the image intervals under F_1, F_2 reduce the length by a scale factor of $\frac{1}{3}$. Length is defined such that for an interval $[a, b]$, $|[a, b]| = b - a$. Then the length of the image intervals are:

$$|F_1([a, b])| = |F_2([a, b])| = \frac{2a + b}{3} - \frac{3a}{3} = \frac{b - a}{3} = \frac{|[a, b]|}{3}$$

Therefore, for all intervals $[a_n, b_n] \in C_n$:

$$|[a_n, b_n]| = 3^{-n}$$

□

Lemma 3. *The Cantor set, $C \in \mathbb{R}$, is compact.*

Proof. The initial state of the Cantor sequence is a closed and bounded interval $[0, 1]$ which is known to be compact from the Heine-Borel theorem. Applying F_1, F_2 to this interval applies translation and scaling which are homeomorphisms and preserve compactness. Therefore, all elements of the Cantor sequence are compact and the intersection of compact sets will also be compact.

□

Proposition 2. *The Cantor set is self similar.*

Proof. Consider the Cantor set C and the image of C under F_1, F_2 . By definition 2 and Proposition 1:

$$F_1(C) \cup F_2(C) = F_1\left(\bigcap_{n=0}^{\infty} C_n\right) \cup F_2\left(\bigcap_{n=0}^{\infty} C_n\right) = \bigcap_{n=0}^{\infty} (F_1(C_n) \cup F_2(C_n)) = \bigcap_{n=0}^{\infty} C_{n+1} = C$$

□

2 Measure Theory

In this section, we introduce methods of measuring sets and fractals which are necessary for this project. For this report, the complement of a set E will be written as E^C and the power set of a space $\mathcal{X}$ will be written as $\mathcal{P}(\mathcal{X})$. Throughout this section, we follow exposition by Gerald Folland (7, pp. 21-32).

2.1 Fields and σ -Fields

Definition 3. *A field of sets on a non-empty underlying set $\mathcal{X}$ is a non-empty collection of subsets, $\mathcal{A}$, on $\mathcal{X}$ which are closed under complements, with respect to the underlying set, and finite unions. This is to say that for all $E_1, E_2, \dots, E_n \in \mathcal{A}$:*

$$\left(\bigcup_{i=1}^k E_i\right), \left(\bigcup_{i=1}^k E_i\right)^C \in \mathcal{A}$$

Definition 4. A σ -field is a field which is closed under countable unions.

$$\forall E_1, E_2, \dots \in \mathcal{A}, \left(\bigcup_{i=1}^{\infty} E_i \right) \in \mathcal{A}$$

The distinction is necessary because of fields which do not satisfy the definition of a σ -field. An example, as given in Example 2.2 of Billingsley's "Probability and Measure" (1), is the field constructed by all intervals $\mathcal{A} = \{(a, b] : 0 \leq a \leq b \leq 1\}$ in the underlying space $(0, 1]$. It is seen that the complement of any set is $(a, b]^C = (0, a] \cup (b, 1]$, which are both elements of the field. The empty set is contained within the field as the set $\emptyset = (a, a]$ and the space itself $(0, 1]$ is also contained. Finally, finite unions of sets will still be composed of elements of the set.

This, however, is not a σ -field. Construct a sequence $b_n = 1 - \frac{1}{n}$ and define the sequence of sets $I_n = (a, b_n]$. The countable union of these sets is $\bigcup_{n=1}^{\infty} I_n = (0, 1) \notin \mathcal{A}$.

Observation 1. An intersection of subsets can be written with a combination of unions and complements.

$$\bigcap_{i=1}^n E_i = \left(\bigcup_{i=1}^n E_i^C \right)^C$$

As a result of this observation, a field is also closed under finite intersections, and a σ -field is closed under countable intersections.

Proposition 3. All fields of an underlying set contain the empty set and the underlying set itself.

Proof. Choose a non empty underlying set $\mathcal{X}$ and a field $\mathcal{A}$. A field is non empty, so there exists an arbitrary set in the field $E \in \mathcal{A}$.

From this, the empty set, $\emptyset$, can be written as $\emptyset = E \cap E^C$. The same can be done for the underlying set itself $\mathcal{X} = E \cup E^C$. $\square$

The definition of a field remains vague with this definition due to the lack of information about how the resulting collection of subsets, $\mathcal{A}$, is constructed. Generally, fields are said to be "generated" from an initial collection of subsets of the underlying set.

Definition 5. A σ -field on an underlying set, $\mathcal{X}$, is said to be generated by a collection of subsets $\mathcal{E} \subseteq \mathcal{P}(\mathcal{X})$ if it is the smallest σ -field on $\mathcal{X}$ containing $\mathcal{E}$. The σ -field generated by $\mathcal{E}$ is denoted $\mathcal{M}(\mathcal{E})$ and is defined as such:

$$\mathcal{M}(\mathcal{E}) = \bigcap \{ \mathcal{M} : \mathcal{M} \text{ is a } \sigma\text{-field on } \mathcal{X}, \mathcal{E} \subseteq \mathcal{M} \}$$

The generation of fields allows for greater understanding of its properties. For example, consider a topological space of $\mathcal{X} = \mathbb{R}$ and a generating subset $\mathcal{E} = [0, 1]$. It is easily seen from Proposition 2 that the collection of subsets in the σ -field is

$$\mathcal{M}([0, 1]) = \{[0, 1], (-\infty, 0) \cup (1, \infty), \emptyset, \mathbb{R}\}$$

Definition 6. A Borel σ -field, $\mathcal{B}_{\mathcal{X}}$, is the σ -field generated by the collection of all open sets in $\mathcal{X}$. All element of $\mathcal{B}_{\mathcal{X}}$ are called Borel sets.

Borel σ -fields are an important structure because they are invariant under a large class of maps.

Lemma 4. *All homeomorphic images of Borel sets are also Borel sets.*

Proof. Consider a Borel set E in a Borel σ -field of an underlying set $\mathcal{X}$. All Borel sets are some countable composition of open and closed maps, as defined in Definition 6.

It is a topological result that the image under a homeomorphism $T : \mathcal{X} \rightarrow \mathcal{X}$ will preserve the preimage's openness or closedness. As a result, the image of the Borel set $T(E)$ is constructed by a countable composition of homeomorphic images of open and closed sets, so $T(E)$ must also be a Borel set. □

Observation 2. *The Cantor set is an element of the Borel σ -field.*

Proof. Recall that the Cantor set is constructed via the Cantor sequence, starting with a closed interval $[0, 1]$ and iteratively applying transformations F_1, F_2 . These functions are homeomorphic because they are composed purely of translation and scaling, leaving the properties of the set it is applied to unchanged.

The initial term of the Cantor sequence is a Borel set because it is closed. Furthermore, as a result of Lemma 4, all elements of the Cantor sequence generated by the homeomorphic maps F_1, F_2 are themselves Borel sets. The Cantor set is defined to be the countable intersection of all elements in the Cantor sequence. It is a result of Observation 2 that a σ -field is closed under countable intersections, therefore the Cantor set is also an element of the Borel σ -field. □

This is not the case for all σ -fields. For example, consider the earlier example for the σ -field of the underlying space $\mathbb{R}$ generated by the interval $[0, 1]$, denoted $\mathcal{M}([0, 1])$. Applying a homeomorphic map defined on $\mathbb{R}$ such as $T(x) = x + 1$ results in the image of the generating interval $T([0, 1]) = [1, 2] \notin \mathcal{M}([0, 1])$.

Proposition 4. *For the Borel σ -field of the real space, $\mathcal{B}_{\mathbb{R}}$, the generating collection of subsets $\mathcal{E} \subseteq \mathcal{P}(\mathbb{R})$, defined such that $\mathcal{M}(\mathcal{E}) = \mathcal{B}_{\mathbb{R}}$, is given to be the collection*

$$\mathcal{E} = \{(a, b) : a < b\}.$$

Proof. All intervals $(a, b) \in \mathbb{R}$ are open, and we know that $(a, b)^C$ will be closed. Trivially, this satisfies the definition for the Borel σ -field. □

2.2 Measures

In order to introduce the important notion of Hausdorff dimension for this project, we must define measure first.

Definition 7. *Let $\mathcal{X}$ be a space with a σ -field $\mathcal{M}$. A measure on $(\mathcal{X}, \mathcal{M})$ is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ which satisfies the below properties. Let $\{E_i\}_1^\infty$ be a sequence of disjoint elements in $\mathcal{M}$:*

- $\mu(\emptyset) = 0$
- $\mu(\bigcup E_i) = \sum(\mu(E_i))$

Definition 8. *For an underlying set $\mathcal{X}$ with a σ -field $\mathcal{M} \subseteq \mathcal{P}(\mathcal{X})$, the space $(\mathcal{X}, \mathcal{M})$ is called a measurable set if there exists an applicable measure on the space. Given a measure μ on this space that satisfies the conditions presented in Definition 7, the space $(\mathcal{X}, \mathcal{M}, \mu)$ called a measure space. A set with a metric*

It is a continuation of this definition that if μ is a measure on a Borel σ -field, then all homeomorphic images of Borel sets are also measurable via μ .

2.2.1 Outer and Complete Measures

Definition 9. An outer measure on a non-empty underlying set $\mathcal{X}$ is a function $\mu^* : \mathcal{P}(\mathcal{X}) \rightarrow [0, \infty]$ which satisfies the following properties:

- $\mu^*(\emptyset) = 0$
- $\mu^*(A) \leq \mu^*(B)$ if $A \subseteq B \subseteq \mathcal{X}$
- If $\{E_i\}_1^\infty \subset \mathcal{X}$ is a collection of disjoint sets, then $\mu^*(\bigcup_1^\infty E_i) \leq \sum_1^\infty \mu^*(E_i)$

The easiest way to construct an outer measure is to consider a collection, $\mathcal{E}$, of elementary sets which countably cover the underlying space $\mathcal{X}$. There must also be a notion of measure, $\rho : \mathcal{E} \rightarrow [0, \infty]$, defined for all elementary sets such that the domain is consistent with other measures, yet does not require additivity. Formally, this may be written as:

Lemma 5. Let $\mathcal{E} \subseteq \mathcal{P}(\mathcal{X})$ and $\rho : \mathcal{E} \rightarrow [0, \infty]$ such that $\emptyset, \mathcal{X} \in \mathcal{E}$ and $\rho(\emptyset) = 0$. Then, for any $E \subset \mathcal{X}$, define

$$\mu^*(E) = \inf \left\{ \sum_1^\infty \rho(A_i) : A_i \in \mathcal{E}, E \subset \bigcup_1^\infty A_i \right\}.$$

Then μ^* is an outer measure.

Proof. It is trivial to see that $\mu^*(\emptyset) = 0$ by letting all $A_i = \emptyset$ and using that $\rho(\emptyset) = 0$. The result that $\mu^*(A) \leq \mu^*(B)$ if $A \subset B$ is also clear because the minimal cover of B must also be a cover of A . Therefore, the focus must only be on the last condition of subadditivity.

To prove the final condition, assume that the set E can be decomposed into a countable union of sets $E = \bigcup_1^\infty E_i \subset \mathcal{P}(\mathcal{X})$, of which each E_i has a countable cover via elementary sets $\{A_i^k\}_{k=1}^\infty \subset \mathcal{E}$ such that $E_i \subset \bigcup_{k=1}^\infty A_i^k$. Choose these covering elementary sets A_i such that for $\epsilon > 0$, it is the case that

$$\sum_{k=1}^\infty \rho(A_i^k) \leq \mu^*(E_i) + \epsilon \cdot 2^{-i}.$$

For the measure of E , $\mu^*(E)$, we are also summing over all i values. This results in the left hand side becoming $\sum_{i,k} \rho(A_i^k)$, which must be greater than or equal to the actual measure, which is the infimum of this summation. Therefore, the inequality becomes

$$\mu^*(E) = \mu^*\left(\bigcup_{i=1}^\infty E_i\right) \leq \sum_{i,k=1}^\infty \rho(A_i^k) \leq \sum_{i=1}^\infty \mu^*(E_i) + \epsilon.$$

Since ϵ is arbitrary, this condition has been proven. $\square$

An outer measure takes a measure on an outer cover of the set. If it is to be a complete measure, ensuring the accuracy of the measure, it must also satisfy the Carathéodory criterion (8, p. 44).

Definition 10. The Carathéodory criterion states that a set $E \in \mathcal{X}$ is μ^* -measurable if the below equality holds for any $A \subseteq \mathcal{X}$:

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^C)$$

The collection, $\mathcal{M}$, of all μ^* -measurable sets is a σ -field and the restriction of μ^* to $\mathcal{M}$ is a complete measure.

While it is not yet necessary, it is of use to prepare the definition for metrics with definitions obtained from Rudin (22, pp. 30-31).

Definition 11. *A metric describes the distance between two points in a geometric space. A space $\mathcal{X}$ equipped with a metric d , written as $(\mathcal{X}, d)$ is a metric space if it satisfies the below conditions for all $x, y \in \mathcal{X}$:*

- $d(x, y) \geq 0$
- $d(x, y) = d(y, x)$
- $d(x, y) = 0 \iff x = y$
- $d(x, z) \leq d(x, y) + d(y, z)$

$(\mathcal{X}, d)$ is a complete metric space if it is also the case that all Cauchy sequences in $\mathcal{X}$ have their limit contained within $\mathcal{X}$.

An example of a metric would be simply the Euclidean metric $d(x, y) = |x - y|$, which trivially satisfies all the above conditions, then $(\mathbb{R}, d)$ is a metric space.

2.3 Hausdorff Measure

The following introduction of Hausdorff measure and, later, dimension follows the exposition of Falconer's "Fractal Geometry" (6, pp. 27-38)

Definition 12. *Define the diameter, $\text{diam} : \mathcal{X} \rightarrow [0, \infty]$ of a set E as*

$$\text{diam}(E) = \sup\{d(x, y) : x, y \in E\}, \quad \text{diam}(\emptyset) = 0.$$

Definition 13. *For $\delta > 0, s \geq 0$, the Hausdorff δ -approximate measure is achieved by covering a set with balls of diameter less than a set value of δ . It is formally defined as*

$$\mathcal{H}_\delta^s(E) = \inf \left\{ \sum_{i=1}^{\infty} \text{diam}(B_i) : B_i \in \mathcal{E}, E \subseteq \bigcup_{i=1}^{\infty} B_i, \text{diam}(B_i) \leq \delta \right\}$$

Proposition 5. *The δ -approximate Hausdorff measure is an outer measure.*

Proof. Simply observe that the form of the δ -approximate Hausdorff measure is the same as in Lemma 5. Thus, it is only necessary to show that $\text{diam}(E)$ is a notion of measure. Recall that a notion of measure has the same domain and value when applied to $\emptyset$ as a standard measure, as well as being defined for all sets in $\mathcal{E}$. As defined in Definition 11, $\text{diam}(\emptyset) = 0$, and the diameter has domain of $[0, \infty]$ for all possible sets. All sets in $\mathcal{E}$ are open balls and the diameter is obviously defined for all balls B_i such that, if the ball B_i is rewritten in terms of its radius r_i , the diameter of the ball is $\text{diam}(B_i) = 2r_i$. $\square$

Definition 14. *The Hausdorff outer measure is a refinement of the δ -approximate Hausdorff measure, taking the limit as δ approaches zero.*

$$\mathcal{H}^{s*} = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^s$$

Lemma 6. *The Hausdorff outer measure satisfies the Carathéodory criterion for the Borel σ -field.*

Proof. By definition, the Borel σ -field is generated by all countable unions and complements of open sets. Because the complement of open sets are closed sets, the Borel σ -field can also be generated by all countable unions and complements of closed sets. As a result, it is suitable to show that if any closed set satisfies the Carathéodory criterion, then the Hausdorff outer measure on the Borel σ -field is a complete measure.

Choose a closed set E and a test set A , then it is a result of countable subadditivity that

$$\mathcal{H}^{s*}(A) \leq \mathcal{H}^{s*}(A \cap E) + \mathcal{H}^{s*}(A \cap E^C)$$

If it is desired to prove the Carathéodory criterion and the equality of the above inequality, it is sufficient to show the opposite inequality. Since E is a closed set, E^C is an open set with some distance to E . Consider a sequence to represent this distance:

$$D_n = \left\{ x \in A \cap E^C : d(x, E) \geq \frac{1}{n} \right\}$$

Taking the limit as $n \rightarrow \infty$, D_n simply approaches $A \cap E^C$. Because the distance $d(D_n, A \cap E) \geq 0$, it is possible to create a system of inequalities for all n :

$$\mathcal{H}^{s*}(A) \geq \mathcal{H}^{s*}(A \cap E) + \mathcal{H}^{s*}(D_n)$$

The limit as $n \rightarrow \infty$ requires substituting in the value for $\lim_{n \rightarrow \infty} D_n = A \cap E^C$:

$$\mathcal{H}^{s*}(A) \geq \mathcal{H}^{s*}(A \cap E) + \mathcal{H}^{s*}(A \cap E^C)$$

As a result of this inequality and the inequality from subadditivity of outer measures, it is shown that equality holds between the two sides and that the Carathéodory criterion holds for all closed sets E . It is also the case that it holds for all countable unions, intersections and complements of the sets. Therefore, the Hausdorff outer measure is a complete measure in the Borel σ -field and all Borel sets are Hausdorff-measurable. □

Proposition 6. *Let there be a set $A \subset \mathbb{R}^n$ and a similarity transformation S of scale factor $\lambda > 0$, then*

$$\mathcal{H}^s(S(A)) = \lambda^s \mathcal{H}^s(A).$$

Proof. Begin with the δ -approximate Hausdorff dimension. If the collection of sets $\{B_i\}$ is a δ -cover of A , then $\{S(B_i)\}$ is a $\lambda\delta$ -cover of $S(A)$. Then observe the inequality between diameters of the ball covers:

$$\sum \text{diam}(S(B_i)) = \lambda^s \sum \text{diam}(B_i)$$

This implies that, after taking the infimum of both δ -approximate Hausdorff measures:

$$\mathcal{H}_{\lambda\delta}^s(S(A)) \leq \lambda^s \mathcal{H}_{\delta}^s(A)$$

Simply letting $\delta \rightarrow 0$ returns the desired inequality. □

2.4 Hausdorff Dimension

It is by definition that a ball B_i in a δ -cover of a set has $\text{diam}(B_i) \leq \delta$. Furthermore, observe that for any set $A \in \mathbb{R}^n$, $0 < \delta < 1$, $0 < s < t \in \mathbb{R}$ and $\{B_i\}$ a δ -cover of A , then

$$\sum \text{diam}(B_i)^t \leq \sum \text{diam}(B_i)^{t-s} \text{diam}(B_i)^s \leq \delta^{t-s} \sum \text{diam}(B_i)^s.$$

This can be applied to show that $\mathcal{H}^t(A) \leq \delta^{t-s} \mathcal{H}^s(A)$ and that the Hausdorff dimension is non-increasing with s . Letting $\delta \rightarrow 0$, we see that if $\mathcal{H}^s(A) < \infty$, then $\mathcal{H}^t(A) = 0$. This also follows the other direction such that if $\mathcal{H}^t(A) > 0$, then $\mathcal{H}^s(A) = \infty$. Using this, it is clear that there exists a critical value of s such that all measures above or below are equal to ∞ and 0 respectively.

Definition 15. *The Hausdorff dimension $\dim_H(A)$ is the critical value of the exponent of the Hausdorff measure. It is formally defined to be:*

$$\dim_H(A) = \inf\{s \geq 0 : \mathcal{H}^s(A) = 0\} = \sup\{s : \mathcal{H}^s(A) = \infty\}$$

The value of the Hausdorff measure of a set A with exponent $s = \dim_H(A)$ can be 0, ∞ , or some finite value between the two. When a set returns $0 < \mathcal{H}^{\dim_H(A)}(A) < \infty$, then it is referred to as an s -set and is easily analysed.

Lemma 7. *A set is an s -set if it is self similar and satisfies the Open Set Condition.*

Proof. see Section 5.3, Lemma 1 of Hutchinson's "Fractals and Self-Similarity" (9, p. 735-736) $\square$

An example of the utility of s -sets is to determine the Hausdorff dimension of the Cantor set, which is known to be an s -set because it is self similar (Proposition 2) and satisfies the Open Set Condition (Lemma 1 and Theorem 1).

The Cantor set can be written as $C = F_1(C) \cup F_2(C)$, where $F_1(C) = \frac{C}{3}$ and $F_2(C) = \frac{C+2}{3}$. It follows that

$$\mathcal{H}^s(C) = \mathcal{H}^s(F_1(C)) + \mathcal{H}^s(F_2(C)).$$

Using Proposition 6, this can be rewritten:

$$\mathcal{H}^s(C) = \frac{1}{3^s} \mathcal{H}^s(C) + \frac{1}{3^s} \mathcal{H}^s(C) = \mathcal{H}^s(C) \cdot \frac{2}{3^s}$$

To solve for the Hausdorff dimension, let s be the Hausdorff dimension of C : $s = \dim_H(C)$. Because C is an s -set, the Hausdorff measure at this value of exponent is known to be finite. This allows the division of both sides by $\mathcal{H}^s(C)$ to be well-defined, following that $3^s = 2$. Therefore, the Hausdorff dimension is solved to be

$$\dim_H(C) = s = \frac{\ln 2}{\ln 3}.$$

Below are the properties satisfied by Hausdorff dimension:

- If $A \subset B$, then $\dim_H(A) \leq \dim_H(B)$.
- For a countable sequence of sets A_i , $\dim_H(\bigcup A_i) = \sup\{\dim_H(A_i)\}$
- If a set $A \subset \mathbb{R}^n$ has non-empty interior $\text{int}(A)$, then $\dim_H(A) = n$.

3 Möbius Maps

Definition 16. Möbius maps are a group of maps with complex variables $m_i \in \mathbb{C}$ defined as

$$M(z) = \frac{m_1 z + m_2}{m_3 z + m_4}.$$

Observation 3. Applying the same scale factor, x , to all variables in a Möbius map, M , leaves the image invariant.

Lemma 8. The group of Möbius maps are constructed by basic transformations of translation, rotation, scaling and inversion on $\hat{\mathbb{C}}$.

Proof. To prove this, it is necessary to have a starting value z and apply only the above transformations to get the Möbius map. We shall re-write the map such that the construction is obvious, starting by removing all z from the numerator:

$$M(z) = \frac{m_1 z + m_2}{m_3 z + m_4} = \frac{m_1(z + m_4/m_3) - m_1 m_4/m_3 + m_2}{m_3(z + m_4/m_3)}$$

Separate the fraction into individual terms:

$$M(z) = \frac{m_1(z + m_4/m_3)}{m_3(z + m_4/m_3)} + \frac{m_2 - m_1 m_4/m_3}{m_3 z + m_4} = \frac{m_1}{m_3} + \frac{m_2 - m_1 m_4/m_3}{m_3 z + m_4}$$

The construction via transformations is then trivial as follows:

$$z \mapsto m_3 z \mapsto m_3 z + m_4 \mapsto \frac{1}{m_3 z + m_4} \mapsto \frac{(m_2 - m_1 m_4/m_3)}{m_3 z + m_4} \mapsto \frac{m_1}{m_3} + \frac{m_2 - m_1 m_4/m_3}{m_3 z + m_4} \equiv \frac{m_1 z + m_2}{m_3 z + m_4}$$

□

A collection of maps which describes geometric transformations such as in Möbius maps are the class of special linear matrices $SL(2, \mathbb{C})$. It is of use to construct an identification map explicitly between the set of Möbius maps and $SL(2, \mathbb{C})$.

Definition 17. Let $SL(2, \mathbb{C})$ be the class of all 2×2 complex matrices of determinant 1. Define $\phi : SL(2, \mathbb{C}) \rightarrow \mathcal{M}$ (where $\mathcal{M}$ is the set of Möbius transformations) such that:

$$\phi \begin{pmatrix} m_1 & m_2 \\ m_3 & m_4 \end{pmatrix} = M(z) = \frac{m_1 z + m_2}{m_3 z + m_4}$$

Lemma 9. The identification map ϕ is a surjective group homomorphism.

Proof. Surjectivity is trivial for ϕ , wherein all matrices of complex elements identify a Möbius map. To prove ϕ is a homomorphism, show that for any matrices $A, B \in SL(2, \mathbb{C})$, $\phi(AB) = \phi(A) \circ \phi(B)$.

Let $A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix}$ and $B = \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix}$. The Möbius maps identified by these maps are:

$$\phi(A) = \frac{a_1 z + a_2}{a_3 z + a_4}, \quad \phi(B) = \frac{b_1 z + b_2}{b_3 z + b_4}$$

Then, simply compute both sides of the equality required for a homomorphism.

$$\begin{aligned}\phi(AB) &= \phi \left(\begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix} \right) = \phi \begin{pmatrix} a_1b_1 + a_2b_3 & a_1b_2 + a_2b_4 \\ a_3b_1 + a_4b_3 & a_3b_2 + a_4b_4 \end{pmatrix} \\ \phi \begin{pmatrix} a_1b_1 + a_2b_3 & a_1b_2 + a_2b_4 \\ a_3b_1 + a_4b_3 & a_3b_2 + a_4b_4 \end{pmatrix} &= \frac{(a_1b_1 + a_2b_3)z + (a_1b_2 + a_2b_4)}{(a_3b_1 + a_4b_3)z + (a_3b_2 + a_4b_4)} \\ \phi(A) \circ \phi(B) &= \frac{a_1\phi(B) + a_2}{a_3\phi(B) + a_4} = \frac{a_1 \frac{b_1z+b_2}{b_3z+b_4} + a_2}{a_3 \frac{b_1z+b_2}{b_3z+b_4} + a_4}\end{aligned}$$

This simplifies to be:

$$\phi(A) \circ \phi(B) = \frac{(a_1b_1 + a_2b_3)z + (a_1b_2 + a_2b_4)}{(a_3b_1 + a_4b_3)z + (a_3b_2 + a_4b_4)} = \phi(AB)$$

□

Observe the kernel of the identification map ϕ , such that $\phi(\text{Ker}(\phi)) = z$. This is a group in the domain $SL(2, \mathbb{C})$ of $\{I, -I\}$.

Definition 18. The 2×2 projective special linear group of complex matrices $PSL(2, \mathbb{C})$ is defined by the quotient group

$$PSL(2, \mathbb{C}) = SL(2, \mathbb{C}) / \{I, -I\}$$

This is an important set to consider because it is a result of Observation 3 that the Möbius map with positive and negative variables are equivalent. As a result, Möbius maps can instead be identified in $PSL(2, \mathbb{C})$.

Lemma 10. The identification map $\phi : PSL(2, \mathbb{C}) \rightarrow \mathcal{M}$ is a homeomorphism.

Proof. It has been proven in Lemma 9 that the identification map is a group homomorphism, and later that it has the kernel $\{I, -I\}$. $PSL(2, \mathbb{C})$ is defined to be the quotient space of the domain against the kernel of ϕ . Therefore, ϕ is a group isomorphism on $PSL(2, \mathbb{C})$.

It is trivial to observe that the coefficients of a Möbius map m_i are continuous functions of the matrix entries. It can also be shown that both $PSL(2, \mathbb{C})$ and the set of Möbius maps $\mathcal{M}$ are topological 3-manifolds. Therefore, it follows via the Open Mapping Theorem that the identification map ϕ is a homeomorphism. □

Its property as a homeomorphism is important such as to ensure that when analysing limit sets and convergence of sequences, the convergence of matrices is consistent with the convergence of maps on the geometric space they are applied onto.

3.1 Isometric circles

It is the goal of this project to make fractals from Möbius maps. To achieve this, we must first study the dynamics of these maps.

Observation 4. The derivative of the Möbius map is given to be $(m_3z + m_4)^{-2}$

Proof. Begin with the quotient rule for derivatives applied to $M(z)$ defined by the matrix $\begin{pmatrix} m_1 & m_2 \\ m_3 & m_4 \end{pmatrix}$:

$$M'(z) = \frac{(m_1z + m_2)'(m_3z + m_4) - (m_3z + m_4)'(m_1z + m_2)}{(m_3z + m_4)^2} = \frac{(m_1m_3z - m_1m_3z) + (m_1m_4 - m_3m_2)}{(m_3z + m_4)^2}$$

Use the property that all elements of the $PSL(2, \mathbb{C})$ group have determinant 1, such that $m_1m_4 - m_3m_2 = 1$. Therefore:

$$M'(z) = (m_3z + m_4)^{-2}$$

□

Definition 19. Consider a map ϕ on a complex metric space $\mathcal{X}$. The isometric circle is defined as the circle on $\mathcal{X}$ where $\phi'(z) = 1$. Points on the interior of the isometric circle have $\phi'(z) > 1$ and points on the exterior of the isometric circle have $\phi'(z) < 1$.

These derivatives can be observed as the scaling of the map at a point. For a local neighbourhood of the space with derivatives above 1, the distance between two points in said local neighbourhood increases when the Möbius map is applied. Similarly, derivatives below 1 are contracting and the distance of two nearby points is decreased under the map at that location.

Proposition 7. The isometric circle of a Möbius map defined by the matrix $\begin{pmatrix} m_1 & m_2 \\ m_3 & m_4 \end{pmatrix}$ has radius $\frac{1}{|m_3|}$ and centre $-\frac{m_4}{m_3}$.

Proof. We let the derivative of the Möbius map be equal to 1, then rearrange to obtain the equation for a circle on the complex plane:

$$\frac{1}{(m_3z + m_4)^2} = 1 \implies |m_3z + m_4| = 1$$

$$\left| z - \left(-\frac{m_4}{m_3}\right) \right| = \frac{1}{|m_3|}$$

This is now in the form of the equation for a circle, in which it is seen that values for radius and centre are correct. □

Definition 20. The centre of inversion, or pole, of an inversion map is the point O which is mapped to infinity. It has the property that for any point P being inverted to P' with respect to a circle of inversion of radius r , $OP \cdot OP' = r^2$.

Lemma 11. A given Möbius map has a circle of inversion with centre and radius equivalent to that of the isometric circle.

Proof. The centre of inversion of a Möbius map is the pole, the point which is mapped to infinity. To find the pole z_∞ of a map $M(z)$, we set the denominator to zero, $m_3z_\infty + m_4 = 0$. Solving this gives $z_\infty = -\frac{m_4}{m_3}$, which corresponds exactly to the centre of the isometric circle.

To find the radius, recall from Lemma 8 that a normalized Möbius map (where $m_1m_4 - m_2m_3 = 1$) can be algebraically deconstructed into a sequence of geometric transformations:

$$M(z) = \frac{m_1z + m_2}{m_3z + m_4} = \frac{m_1}{m_3} - \frac{1}{m_3^2 \left(z + \frac{m_4}{m_3}\right)}$$

This expression reveals a four-step geometric construction:

1. Translation to bring the pole to the origin: $z \mapsto z + \frac{m_4}{m_3}$
2. Complex inversion around the origin: $z \mapsto \frac{1}{z}$
3. Dilation and rotation: $z \mapsto -\frac{1}{|m_3|^2}z$
4. Translation back to original pole position: $z \mapsto z + \frac{m_1}{m_3}$

We wish to track the distance r from a point P to the centre of inversion C through these transformations. Translations and rotations leave this radial distance unchanged. Therefore, the only transformations that alter r are the inversion ($r \mapsto \frac{1}{r}$) and the dilation by a factor of $\frac{1}{|m_3|^2}$. Applying these consecutively maps the distance as follows:

$$r \mapsto \frac{1}{|m_3|^2 r}$$

The circle of inversion is defined by the set of points where this radial distance is preserved. Setting the final distance equal to the initial distance gives:

$$r = \frac{1}{|m_3|^2 r} \implies r^2 = \frac{1}{|m_3|^2} \implies r = \frac{1}{|m_3|}$$

□

Definition 21. *The fixed point of a map $\phi : A \rightarrow A$ is the point $a \in A$ such that $\phi(a) = a$.*

Finding the fixed point of the Möbius map is easy to compute. Consider the equation with a generic Möbius map:

$$M(z) = \frac{m_1 z + m_2}{m_3 z + m_4} = z$$

. This can be rearranged to give the quadratic equation $m_3 z^2 + (m_4 - m_1)z - m_2 = 0$. The two fixed points $\gamma_{\pm}$ are the solutions to this equation.

Lemma 12. *A fixed point of a Möbius map M_i exists inside either the isometric circle of M_i or the isometric circle of M_i^{-1} .*

Proof. Consider any fixed point γ of a map $M_i = \begin{pmatrix} m_1 & m_2 \\ m_3 & m_4 \end{pmatrix}$. It is known that the fixed point is a solution of the equation $m_3 \gamma^2 + (m_4 - m_1)\gamma - m_2 = 0$ and that an isometric circle has centre $\frac{-m_4}{m_3}$ and radius $\frac{1}{|m_3|}$. Consider the distance between the two centres of the isometric circles:

$$\left(\gamma + \frac{m_4}{m_3}\right)\left(\gamma - \frac{m_1}{m_3}\right) = \gamma^2 + \frac{m_4 - m_1}{m_3} - \frac{m_1 m_4}{m_3^2}$$

Use the property from rearranging the fixed point equation that $\gamma^2 + \frac{m_4 - m_1}{m_3} \gamma = \frac{m_2}{m_3}$ to give:

$$\left(\gamma + \frac{m_4}{m_3}\right)\left(\gamma - \frac{m_1}{m_3}\right) = \frac{m_2 m_3 - m_1 m_4}{m_3^2} = \frac{-1}{m_3^2}$$

Therefore, when finding the distance between the points, we get that

$$\left|\gamma + \frac{m_4}{m_3}\right| \left|\gamma - \frac{m_1}{m_3}\right| = \frac{1}{|m_3|^2} = r^2.$$

If $|\gamma + \frac{m_4}{m_3}| = |\gamma - \frac{m_1}{m_3}|$, then γ is touching both isometric circles at the tangency point. If $|\gamma + \frac{m_4}{m_3}| \neq |\gamma - \frac{m_1}{m_3}|$, then it must be that one distance is greater than r and one is less than r , implying that the fixed point is inside one isometric circle and outside the other. □

Proposition 8. *A Möbius map with distinct eigenvalues is diagonalisable into a matrix of equal trace and determinant.*

Proof. Consider a general Möbius map defined by a 2×2 matrix, M . There exists two eigenvalues $\lambda_{\pm}$ of this matrix as the solution to $(M - \lambda I)v = 0$, in which we assume that $\lambda_+ \neq \lambda_-$ and we have two distinct eigenvalues. It then follows that there exists two distinct eigenvectors $v_{\pm}$ and an eigenvector matrix $P = (v_+ \ v_-)$ such that $PMP^{-1} = M' = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix}$.

It then follows from simple calculation that their trace is equal:

$$\text{Tr}(PMP^{-1}) = \text{Tr}(MP^{-1}P) = \text{Tr}(M)$$

as well as their determinant:

$$\text{Det}(PMP^{-1}) = \text{Det}(M) \cdot \text{Det}(PP^{-1}) = \text{Det}(M)$$

□

Observation 5. *The eigenvector matrix of a Möbius map with distinct eigenvalues can be transformed into a Möbius map itself.*

Proof. The eigenvector matrix of M is known to be invertible. It also follows that the scale factor of the map is not important for the resulting value of M' , including scaling by a negative value. We may introduce a new map $\hat{P} = \frac{1}{\sqrt{\text{Det}(P)}}P$ such that $PMP^{-1} = \hat{P}M\hat{P}^{-1}$ and $\hat{P} \in PSL(2, \mathbb{C})$, and therefore $\hat{P} \in \mathcal{M}$, the group of Möbius maps. □

It is as a result of all maps constructing the diagonalised matrix M' being Möbius maps that M' must also be a Möbius map. A diagonalised matrix is a lot easier to work with, especially when considering the analysis of how a Möbius map effects the space $\hat{\mathbb{C}}$.

Proposition 9. *For a 2×2 matrix M with a double eigenvalue λ and eigenvector v , a generalised eigenvector w may be found by the equation $(M - \lambda I)w = v$.*

Theorem 2. *A matrix with a double eigenvalue has an eigenvector matrix $P = (v \ w)$ with v the eigenvector and w the generalised eigenvector from the eigenvalue λ . The resulting normalised matrix $M' = PMP^{-1}$ is of the Jordan normal form $M' = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$.*

It is a property of Möbius maps that the determinant of the map is equal to 1. It is therefore as in Proposition 8 that the determinant of M' is also equal to 1. Since the determinant of M' is simply λ^2 , we have the result $\lambda = \pm 1$. For simplicity and because both choices of λ may result in the same dynamics, we may choose $\lambda = 1$ and therefore the normalisation of a Möbius map with double eigenvalue is the canonical parabolic form $M' = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Proposition 10. *The normalised form of a Möbius map maps its fixed points to the points of ∞ and 0, or solely ∞ if the matrix M has double eigenvalue.*

Proof. A normalised non-defective matrix is of the form $\begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix} \in PSL(2, \mathbb{C})$, which identifies a Möbius map $M(z) = \frac{\lambda_+}{\lambda_-}z$. This is a simple scaling transformation which trivially has fixed points at $z = 0$ and $z = \infty$.

The normalised Möbius map of a map with repeated eigenvalues is the canonical parabolic form $M(z) = z + 1$, which trivially has a single fixed point at $z = \infty$. □

The normalised form of a Möbius map can be used to observe the fundamental dynamics of the map in the geometric space $\hat{\mathbb{C}}$. We may state that a generalised map is of either of the forms:

$$M'(z) = \begin{cases} \frac{\lambda_+}{\lambda_-} z, & \text{if } \lambda_+ \neq \lambda_- \\ z + 1, & \text{if } \lambda_+ = \lambda_- \end{cases}$$

With this, we can observe differences in the maps produced as such by considering the constant scaling the z term of the normalised Möbius map, referred to as the multiplier $k = \frac{\lambda_+}{\lambda_-}$.

Proposition 11. *A given multiplier for a normalised Möbius map $k = \frac{\lambda_+}{\lambda_-}$ is related to the trace of the map $\text{Tr}(M)$ by the equation $k + \frac{1}{k} = |\text{Tr}(M)|^2 - 2$.*

Proof. Recall that Möbius maps have determinant of 1, so a normalised map has the property $\lambda_+ \lambda_- = 1$ or $\lambda_+ = \frac{1}{\lambda_-}$. As a result, the multiplier k can be rewritten to be $k = \lambda_+^2$ and, similarly, $\frac{1}{k} = \lambda_-^2$. Therefore,

$$k + \frac{1}{k} = \lambda_+^2 + \lambda_-^2 = (\lambda_+ + \lambda_-)^2 - 2\lambda_+ \lambda_- = \text{Tr}(M)^2 - 2$$

When the Möbius map has repeated eigenvalues $\lambda_+ = \lambda_-$, we may simply check the above condition for the value of $k = 1$ and $\text{Tr}(M) = 2$ to see that $k + \frac{1}{k} = 2 = |2|^2 - 2$. Therefore, this property holds for all Möbius maps. $\square$

We observe, too, that if $\frac{\lambda_+}{\lambda_-}$ has a non-zero imaginary component, this scale factor acts as a rotation upon z . We can then use this information to produce classification for Möbius maps dependent on their multiplier k or, equivalently, their trace $\text{Tr}(M)$.

Theorem 3. *A Möbius map has 4 possible classifications depending on the action of their normalised matrix. The classifications and generic dynamics of the map, denoted by $S(z)$, is given below:*

- *Parabolic:* $k = 1, \text{Tr}(M) = 2, S(z) = z + \alpha$
- *Elliptic:* $|k| = 1, \arg(k) \neq 0 \pmod{2\pi}, |\text{Tr}(M)| < 2, S(z) = e^{i\theta} z$
- *Hyperbolic:* $k > 0, \arg(k) = 0 \pmod{2\pi}, \text{Tr}(M) > 2, S(z) = kz$
- *Loxodromic:* $|k| \neq 1, \arg(k) \neq 0 \pmod{2\pi}, |\text{Tr}(M)| > 2, S(z) = re^{i\theta} z$

To apply the theory and showcase the method for the classification of Möbius maps, recall the maps which are used to iteratively construct the Cantor set $F_1(X) = \frac{X}{3}, F_2(X) = \frac{X+2}{3}$. These are purely composed of geometric transformations, thus we can write these maps as matrices:

$$F_1 = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, F_2 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$$

It is then important to normalise these matrices to ensure that they have unitary determinant. The determinant of both matrices is 3, so apply a scale factor of $\frac{1}{\sqrt{3}}$ to both matrices to obtain the Möbius maps

$$\hat{F}_1 = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & \sqrt{3} \end{pmatrix}, \hat{F}_2 = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{3}} \\ 0 & \sqrt{3} \end{pmatrix}.$$

These Möbius maps are already normalised enough to sufficiently classify them. $\hat{F}_1$ is diagonalised and has a real multiplier $k > 0$, implying that it is a hyperbolic map. $\hat{F}_2$ has a real trace $\text{Tr}(\hat{F}_2) > 2$, implying that it is also a hyperbolic map. Therefore, the Cantor set is constructed with two hyperbolic Möbius maps.

4 Generators

We will now introduce the mathematical structure for the systems which will allow us to make fractal images from Möbius maps.

4.1 Schottky Groups

Proposition 12. *The Riemann sphere $\hat{\mathbb{C}}$ equipped with the chordal metric χ is a complete metric space.*

Proof. The chordal metric is defined to be

$$\chi(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \cdot \sqrt{1 + |w|^2}}.$$

We observe that the space $\hat{\mathbb{C}}$ equipped with the chordal metric χ is isometric to the unit sphere $\mathbb{S}^2$ equipped with the Euclidean metric d via a stereographic projection (17, pp. 139-147). Since $\mathbb{S}^2$ is a compact space, that must also be the case for $\hat{\mathbb{C}}$. Every sequence in a compact metric space has a convergent subsequence with limit L . If the sequence itself is Cauchy, then it must also converge to the limit L . As a result, the metric space $(\hat{\mathbb{C}}, \chi)$ is complete. $\square$

Definition 22. *A classical Schottky group is generated by a finite set of n loxodromic Möbius maps $\{M_1, M_2, \dots, M_n\}$ called generators acting upon the geometric space $\hat{\mathbb{C}}$. The group is defined to satisfy the Schottky condition such that for $2n$ disjoint Schottky disks $D_1, D'_1, D_2, D'_2, \dots, D_n, D'_n \in \hat{\mathbb{C}}$, M_i maps the exterior of D'_i to the interior of D_i . This is written formally to be $M_i(\hat{\mathbb{C}} \setminus D'_i) \subset D_i$. The generators in a Schottky group are contraction maps, meaning that in their domain $\hat{\mathbb{C}} \setminus D'_i$, there exists a contractivity factor $s_n \in [0, 1)$ such that for all $x, y \in (\hat{\mathbb{C}} \setminus D'_i)$ and a generator M_i , $d(x, y) \leq s_n d(M_i(x), M_i(y))$.*

What is of note with this definition is that the map M_i has an isometric circle which defines the boundary of D'_i as known from Lemma 11. With the removal of D'_i from the domain, we ensure that the derivative of M_i is less than 1 for all points. Consequently from Definition 24, the inverse map M_i^{-1} shall map the exterior of D_i into D'_i .

Lemma 13. *Choose any starting compact set $E_0 \in \mathcal{X}$ and a Schottky group. The limit of the iterative sequence $E_{k+1} = \bigcup_{i=1}^n M_i(E_k)$ will converge to the limit set of the Schottky group.*

Proof. The space $\hat{\mathbb{C}}$ equipped with the chordal metric is a complete metric space, meaning that it contains all possible limits of Cauchy sequences in $\mathcal{X}$. The sequence constructed with iterations of M_i must converge in $\mathcal{X}$ because all M_i are contraction maps. Let the sequence converge to the term E_∞ . It then follows that $E_\infty = \bigcup_{i=1}^n F_i(E_\infty)$, which is the definition for the attractor Λ . $\square$

Definition 23. *A kissing Schottky group is an alteration of a Schottky group such that consecutive disks are tangent to each other $D_i \cap D_{i+1} = \gamma_i$. The Möbius maps in the group are defined such that $M_{i+1}(\gamma_i) = \gamma_{i+1}$.*

It is worth also introducing a way to label or describe the images of disks under a Schottky group.

Definition 24. *A word W of length $|W| = n$ corresponds to a composition of n generators. A word is applied from right to left, such that in the word $F_k \dots F_3 F_2 F_1$, F_1 is applied first.*

We may also take compositions of words, such that if there are two words $W = F_i F_2$ and $Q = F_3 F_4$, then $WQ = F_1 F_2 F_3 F_4$. Compositions of words are also applied right to left, so we must first apply Q and then W afterwards.

Definition 25. Words can be associated to each image disk to describe their position and construction in the image. Consider the word $W_n = g_1 \circ g_2 \circ \cdots \circ g_n$, then the image disk associated with this word is referred to as D_W and is constructed by applying Möbius maps to the initial circle as such:

$$D_W = g_1 \circ g_2 \circ \cdots \circ g_{n-1}(D_{g_n})$$

Some examples of words associated to image disks are provided below, where D_g is the Schottky disk associated with the generator g :

- $g_1 = Id(D_{g_1}) = D_{g_1}$
- $C_{g_1 g_2} = g_1(C_{g_2})$
- $D_{g_1 g_1^{-1} g_2} = g_1 g_1^{-1}(D_{g_2}) = Id(D_{g_2}) = D_{g_2}$

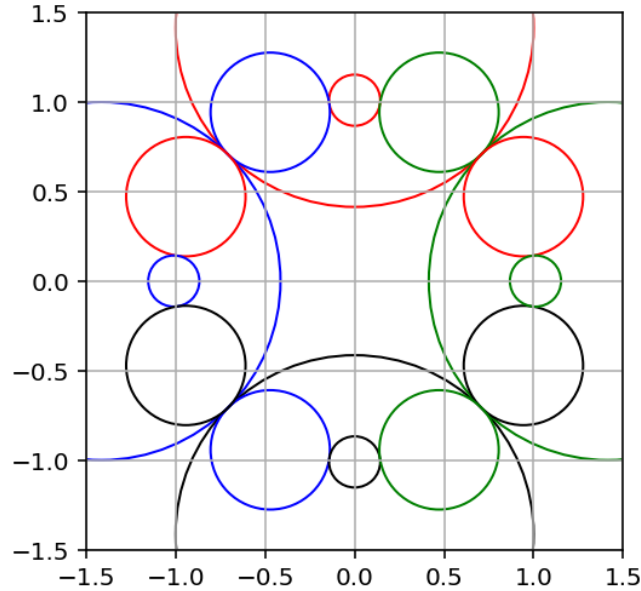


Figure 7: Image displaying isometric and image circles for word lengths up to 2. Circles are coloured with respect to the first generator in the word. a = red, b = blue, A = black, B = green.

Proposition 13. A kissing Schottky group's commutator $[M_1, M_2, \cdots, M_n]$ is parabolic, such that $|Tr([M_1, M_2, \cdots, M_n])| = 2$.

Proof. Consider a tangency point γ_0 which is the first tangency point between D_1 and D'_n , both of which bounded by the isometric circles of M_1 and M_n^{-1} respectively. It is known that $M_1(\gamma_0) = \gamma_1$ and $M_2(\gamma_1) = \gamma_2$. Therefore, it is a suitable continuation to take compositions of maps and see that $M_1 M_2 M_3 \cdots M_n M_1^{-1} \cdots M_n^{-1}(\gamma_0) = [M_1, M_2, \cdots, M_n](\gamma_0) = \gamma_0$. This is by definition the only possible fixed point, thus the only possible classification of the commutator's Möbius map is for it to be parabolic. $\square$

Proposition 14. *A classical Schottky group's commutator is loxodromic, such that*

$$|Tr([M_1, M_2, \dots, M_n])| > 2.$$

Proof. If the commutator W is loxodromic, then there exists two fixed points - an attractive fixed point and a repulsive fixed point. We may find the attractive fixed point by repeatedly applying W to a set and, similarly, we may find the repulsive fixed point by repeatedly applying W^{-1} . First, apply W to the disk D_1 associated with the map M_1 . It is trivial to see that $W(D_1) \subset D_1$, thus by Brouwer's fixed point theorem, the attractive fixed point is contained within D_1 . Similarly, we see that $W^{-1}(D_2) \subset D_2$, therefore the repulsive fixed point is contained within D_2 . We know that the Schottky disks are disjoint as well as that $W(D_i) \cap D_i = \emptyset$, therefore there are two disjoint fixed points and the map is loxodromic. $\square$

4.1.1 Limit Sets

Definition 26. *The limit set generated by a Schottky group is defined to be the collection of points constructed by applying all words of infinite length to any initial compact set in the geometric space, typically chosen to be one of the Schottky disks D_i corresponding to the map M_i . If we define G_n as the set of all elements of the Schottky group with word length greater than n and ensure that the composition of maps is $g = g_1 g_2 \dots g_n : g_i \neq g_{i-1}^{-1}, g_n \neq M_i^{-1}$, the limit set is written formally to be $\Lambda = \bigcap_{n=0}^{\infty} \left(\bigcup_{g \in G_n} g(D_i) \right)$*

We shall note a few features and definitions of limit sets constructed with these Schottky groups to precede future constructions.

Observation 6. *Given a Möbius map M_i in a Schottky group, the fixed point contained within the map's Schottky disk $z_0 \in D_i$ is contained within the limit set generated by the Schottky group.*

Proof. Trivially, the fixed point z_0 of M_i corresponds to the infinite word $\lim_{n \rightarrow \infty} M_i^n$. $\square$

Lemma 14. *Choose any starting compact set $E_0 \in \mathcal{X}$ and a Schottky group. The limit of the iterative sequence $E_{k+1} = \bigcup_{i=1}^n M_i(E_k)$ will converge to the limit set of the Schottky group.*

Proof. The space $\hat{\mathbb{C}}$ equipped with the chordal metric is a complete metric space, meaning that it contains all possible limits of Cauchy sequences in $\mathcal{X}$. The sequence constructed with iterations of M_i must converge in $\mathcal{X}$ because all M_i are contraction maps. Let the sequence converge to the term E_∞ . It then follows that $E_\infty = \bigcup_{i=1}^n M_i(E_\infty)$, which is the definition for the attractor Λ . $\square$

Observation 7. *The limit set constructed by a Schottky group is a Cantor dust.*

Proof. A Cantor dust describes the disconnected points such as seen in the Cantor set. It is observed that for a Schottky group, all disks are disjoint. Under standard dynamics of the Schottky group such that under a map M_i , the exterior of D'_i is mapped to the interior of D_i . It follows from Möbius maps' geometric, and thus homeomorphic, properties that the image of disks under M_i is also disjoint in D_i . It is then trivial to conclude that there must exist gaps between all limit points contained within the limit set, presenting as the gaps between image disks, and that the limit set is a Cantor dust. $\square$

Observation 8. *The limit set constructed via a kissing Schottky group is a quasi-circle.*

Proof. A kissing Schottky group has no gaps between the disks by definition. When infinitely iterating the generators upon a starting compact set to find the limit set, the number of points in a sequence element becomes dense, therefore leaving no gaps in the resulting limit set. The resulting shape is a connected set homeomorphic to $\mathbb{S}^1$, therefore it is a quasi-circle. $\square$

5 Limit Set Types

With the previous sections, we have covered all necessary prerequisites to explore the different types of limit sets and the formulae for how to make the Schottky group which constructs the limit sets in this project. For future reference, we will use a Schottky group generated by 2 2 Möbius maps denoted a and b , with their inverses A and B respectively. The Schottky disks for these maps are D_a, D_b, D_A, D_B such that for any Möbius map (without loss of generality, choose A): $A(\hat{\mathbb{C}} \setminus D_a) \subset D_A$.

This section will be following exposition from the final roadmap at the end of Indra's Pearls to dictate the order and naming for the types of limit sets (16, p. 396). Knowing that Möbius maps may be classified via their trace by Theorem 2 and that traces impact the resulting limit set by Observation 6 and 7, we shall observe the trace of the maps a, b, ab and the commutator $abAB$.

5.1 Classical Schottky Dust

Classical Schottky Dust is the type of limit set constructed by a typical classical Schottky group. As per the definition of a classical Schottky group, loxodromic maps a, b are chosen such that their Schottky disks are disjoint. The resulting limit set leaves gaps between limit points, with similarities to Cantor dust yet in $\hat{\mathbb{C}}$. The conditions on the trace of maps is that

$$|\mathrm{Tr}(a)|, |\mathrm{Tr}(b)|, |\mathrm{Tr}(ab)|, |\mathrm{Tr}(abAB)| > 2.$$

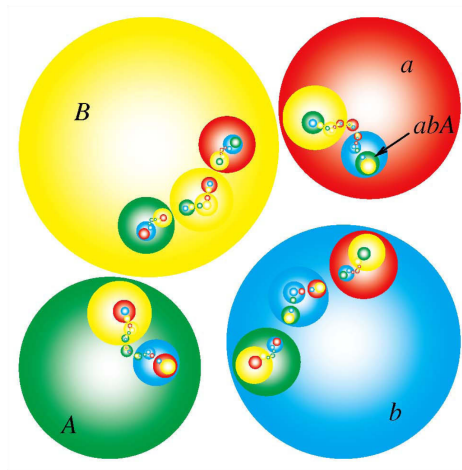


Figure 8: A picture example of the map dynamics of a classical Schottky group which generates a classical schottky dust with labelled Schottky disks, obtained from page 106 of Indra's Pearls (16). It is clear that the resulting limit set will be disconnected by the gaps between these image disks.

5.1.1 Fuchsian Classical Schottky Dust

Definition 27. A Fuchsian limit set is a set which lies entirely on a geometric circle in the Riemann sphere.

It is useful for future analysis to consider what requirements there are for the generators to have a Fuchsian limit set.

Proposition 15. A limit set is Fuchsian if the traces of a, b, ab are all real values.

Proof. On the Riemann sphere, all circles are equivalent. Thus, we may map the geometric circle the limit set lies on to the extended real line $\mathbb{R} \cup \{\infty\}$ using a Möbius map. Recall from Theorem 2 by the classification of maps that maps with complex traces apply a rotation to the space. Therefore, it is necessary to ensure that all traces are real. The fewest necessary cases to consider are the traces of the maps a, b, ab . $\square$

From this, we may then want to define a collection of Fuchsian conditions such that we may use them in the future.

Definition 28. The conditions required to check if a collection of generators returns a Fuchsian limit set are given as Fuchsian conditions:

- $2 < \text{Tr}(a), \text{Tr}(b), \text{Tr}(ab) \in \mathbb{R}$
- $\text{Det}(a), \text{Det}(b) = 1$
- $\text{Tr}(abAB) \geq 2$

It is known from Proposition 13 that the Fuchsian classical Schottky dust occurs when the commutator is loxodromic, thus for this case, we have strictly $\text{Tr}(abAB) > 2$.

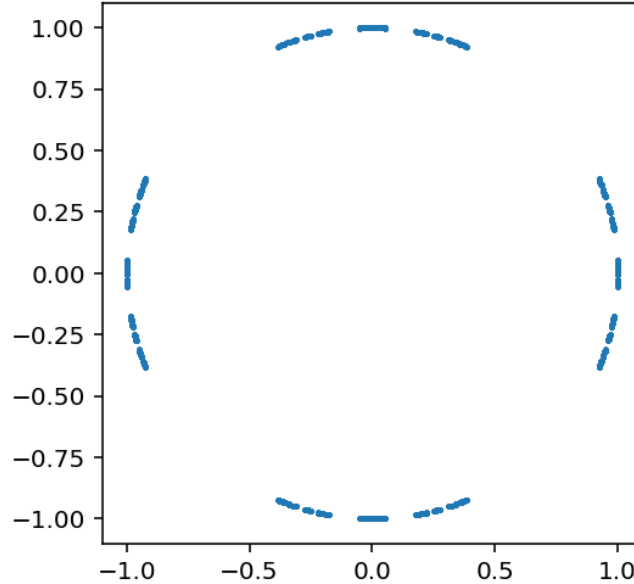


Figure 9: Fuchsian classical Schottky dust fractal constructed with $\theta = 0.7$ using the formulae for generators in Section 6.1.

5.2 Quasi-Circle Limit set

Using a kissing Schottky group, the resulting limit set is known to be connected and, furthermore, homeomorphic to a circle. For a kissing Schottky group, the necessary conditions are that $|\text{Tr}(abAB)| = 2$ and that a, b, ab can have any complex trace of magnitude greater than 2. The resulting fractals are quasicircles, of which their Hausdorff dimension has been studied to be bounded by $1 \leq \dim_H(\lambda) < 2$. When Λ is not the circle and thus has infinite complexity, the dimension is known to be strictly greater than 1 (23).

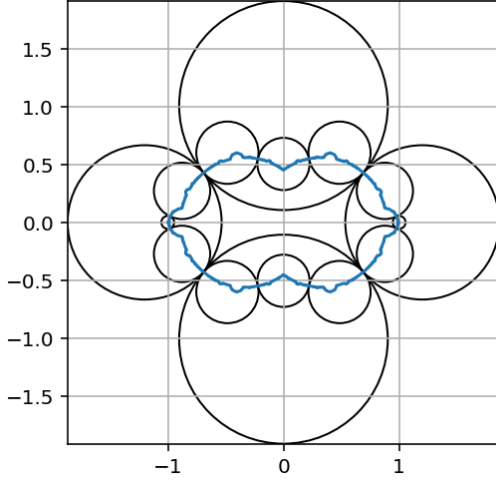


Figure 10: This is a quasi-circle limit set constructed with variables $x = 0.7$, $y = 1$ with formulae given in Section 6.2.

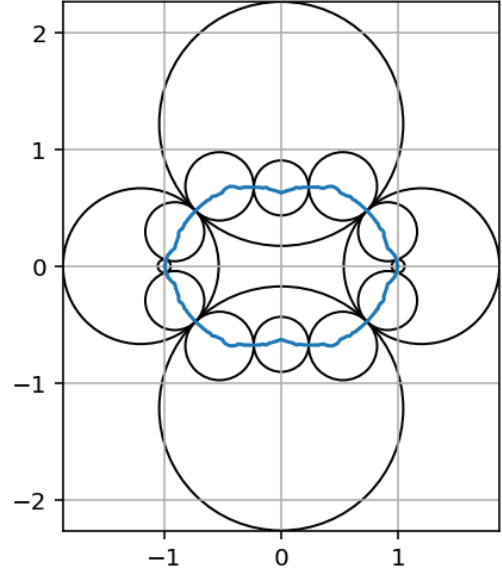


Figure 11: Quasi-circle limit set constructed with two variables $x = 0.6$, $y = 1.5$ with formulae given in Section 6.2.

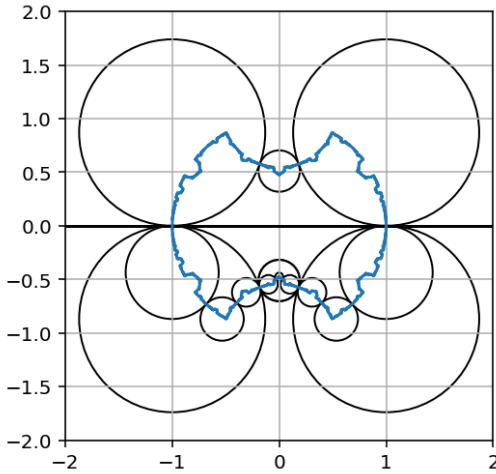


Figure 12: This is a quasi-circle limit set constructed with variables $\text{Tr}(a) = 2.7$, $\text{Tr}(b) = 2.3$ with formulae given in Section 6.3.

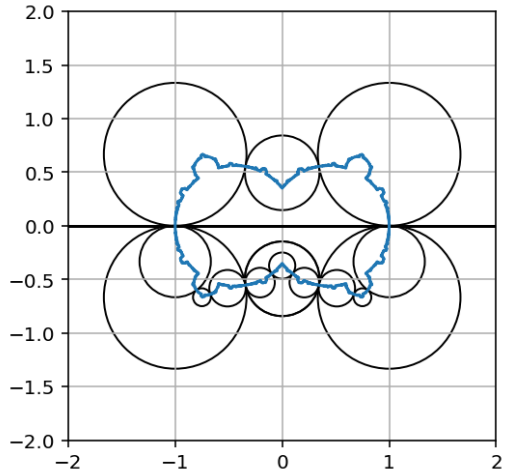


Figure 13: Quasi-circle limit set constructed with two variables $\text{Tr}(a) = 2.2$, $\text{Tr}(b) = 3$ with formulae given in Section 6.3.

5.2.1 Fuchsian Connected Limit Set

A Fuchsian connected limit set is a limit set which is homeomorphic to a circle and with all limit points led upon a geometric circle on $\hat{\mathbb{C}}$. This results in the limit set simply being the geometric circle and the Hausdorff dimension of the limit set being $\dim_H(\Lambda) = 1$. We see that this occurs when the Fuchsian conditions defined earlier are satisfied, as well as requiring a parabolic commutator as per the definition of kissing Schottky groups.

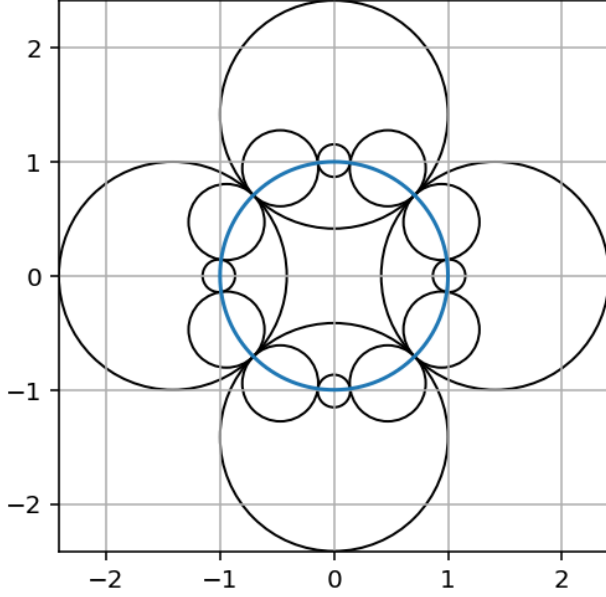


Figure 14: This is a Fuchsian connected limit set constructed with variables $x = 1, y = 1$ with formulae provided in Section 6.2.

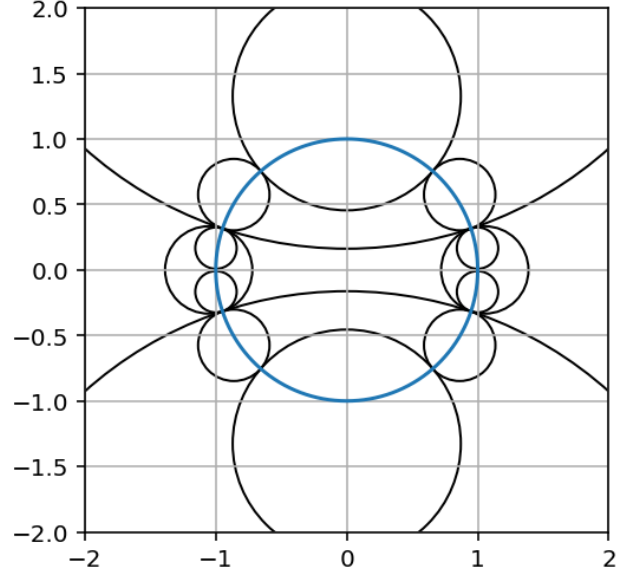


Figure 15: This is a Fuchsian connected limit set constructed with variables $x = \frac{1}{3}, y = 3$ with formulae provided in Section 6.2.

5.3 Single Cusp Groups

Construct a kissing Schottky group such that all maps have absolute value of trace at least 2 and the commutator is parabolic, with the condition that one of either a or b is a parabolic Möbius map with absolute value of trace equal to 2.

Observation 9. *Given a parabolic Möbius map, its fixed point and the centre of its Schottky disk are the same point.*

Proof. Using the normalisation of parabolic Möbius maps, we know that they may be written in canonical parabolic form $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. We shall use Proposition 7 to state that the centre of this Möbius map is $\frac{1}{0}$ which, in the extended complex plane of the Riemann sphere, is equal to infinity. We also recall that our normalised parabolic Möbius map has a single fixed point at the point of infinity. Therefore, the centre and fixed points are the same point and any Möbius map applied to the map will preserve this property. $\square$

We may observe the interesting nature of the single cusp groups by considering the collection of limit set images as the trace of the map b tends towards 2.

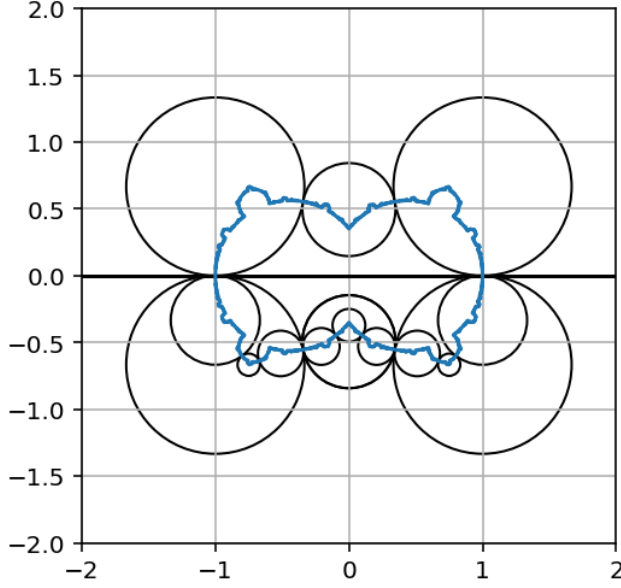


Figure 16: Kissing Schottky group with traces $\text{Tr}(a) = 2.2$, $\text{Tr}(b) = 3$. Formulae for generators found in Section 6.3.

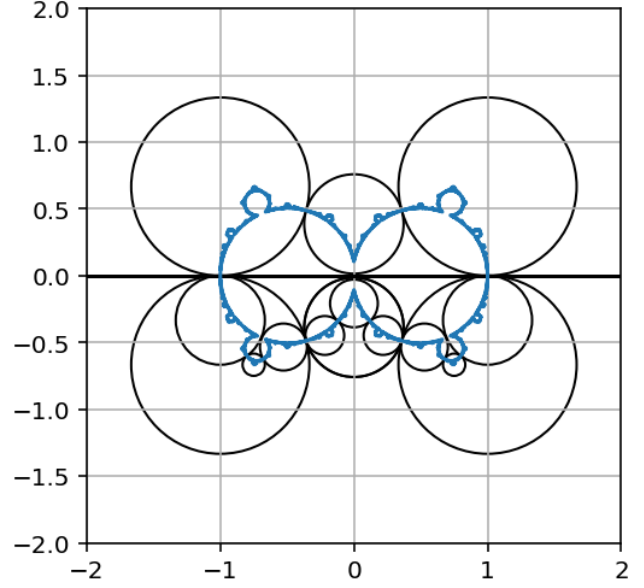


Figure 17: Kissing Schottky group with traces $\text{Tr}(a) = 2.02$, $\text{Tr}(b) = 3$. Formulae for generators found in Section 6.3.

What is observed is that the fixed points of the map a , and thus also the limit points associated with the infinite application of the map a or A as per Lemma 14, eventually approach each other until they touch upon the map being parabolic. The resulting limit set is connected, however not simply connected due to the countable connected sections which take the appearance of bulbs between all combinations of words ending in an infinite string of the parabolic map a and A .

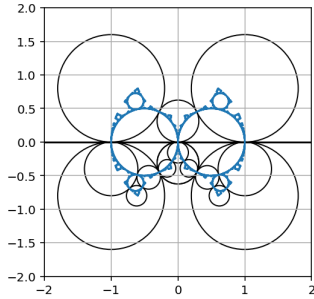


Figure 18: Kissing Schottky group with traces $\text{Tr}(a) = 2$, $\text{Tr}(b) = 2.5$. Formulae for generators found in Section 6.3.

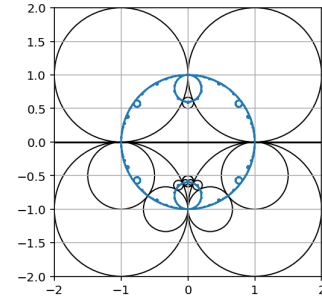


Figure 19: Kissing Schottky group with traces $\text{Tr}(a) = 4$, $\text{Tr}(b) = 2$. Formulae for generators found in Section 6.3.

5.4 Double Cusp Groups

Double cusp groups are groups in which two Möbius maps (excluding the commutator) in the Schottky group are parabolic, as well as the commutator being parabolic.

Lemma 15. *The traces of the maps a, b, ab produce the following equality for kissing Schottky groups:*

$$\text{Tr}(a)^2 + \text{Tr}(b)^2 + \text{Tr}(ab)^2 = \text{Tr}(a) \text{Tr}(b) \text{Tr}(ab) + 4$$

Proof. We shall use the Cayley-Hamilton theorem for 2×2 matrices of determinant 1:

$$\text{Tr}(M)\text{Tr}(N) = \text{Tr}(MN) + \text{Tr}(MN^{-1})$$

. Begin by reformatting the commutator trace $\text{Tr}(abAB) = \text{Tr}(a \cdot bAB)$. We may then rearrange and apply the Cayley-Hamilton theorem:

$$\text{Tr}(a \cdot bAB) = \text{Tr}(a)\text{Tr}(bAB) - \text{Tr}(A(bAB)^{-1}) = \text{Tr}(a)^2 - \text{Tr}(ABAb)$$

Then, we can apply the Cayley-Hamilton theorem again to $\text{Tr}(AB \cdot Ab)$:

$$\text{Tr}(ABAb) = \text{Tr}(AB)\text{Tr}(Ab) - \text{Tr}(ABBa)$$

. Finally, use the identity to show $\text{Tr}(Ab) = \text{Tr}(a)\text{Tr}(b) - \text{Tr}(ab)$, as well as $\text{Tr}(AB) = \text{Tr}(ab)$ and $\text{Tr}(ABBa) = \text{Tr}(B^2) = \text{Tr}(b)^2 - \text{Tr}(I) = \text{Tr}(b)^2 - 2$. This gives the final equality

$$\text{Tr}(abAB) = \text{Tr}(a)^2 + (\text{Tr}(b)^2 - 2) - \text{Tr}(ab)[\text{Tr}(a)\text{Tr}(b) - \text{Tr}(ab)]$$

$$\text{Tr}(abAB) = \text{Tr}(a)^2 + \text{Tr}(b)^2 + \text{Tr}(ab)^2 - \text{Tr}(a)\text{Tr}(b)\text{Tr}(ab) - 2$$

Rearranging the equation and using the kissing Schottky property that $\text{Tr}(abAB) = 2$ gives the required solution $\square$

As a result, we may solve for a parabolic generator which is a combination of generators a, b by first constructing its trace in terms of $\text{Tr}(a), \text{Tr}(b)$ using the above Markov identity, then simply solving the resulting polynomial equation. In the simple case that we just let a, b both be parabolic, the resulting limit set is the Apollonian gasket of dimension $\dim_H(\Lambda) \approx 1.3056$ (24).

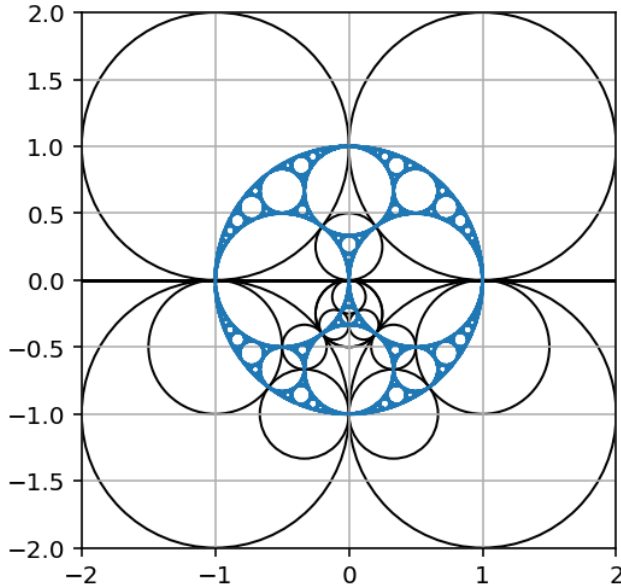


Figure 20: The Apollonian gasket constructed with $\text{Tr}(a) = \text{Tr}(b) = 2$ for the generator formulae in Section 6.3.

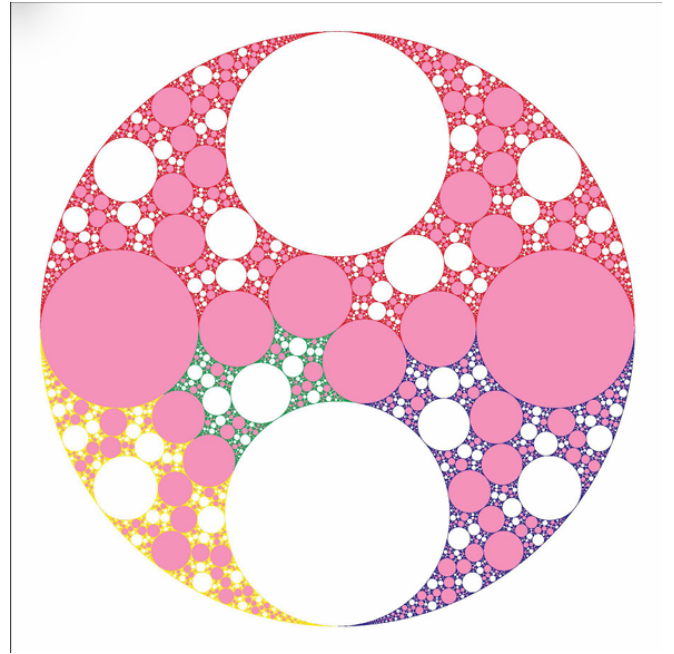


Figure 21: Kissing Schottky group with traces $\text{Tr}(a) = 1.64213876 - 0.76658841i$, $\text{Tr}(b) = 2$. Formulae for generators found in Section 6.3. The trace of a is chosen such that the word a^3Ba^2B is parabolic. Limit points are coloured depending on the first letter of their word, with red for a , blue for b , green for A , and yellow for B . Image source: (16, p.272)

6 Schottky Group Generator Formulae

Knowing the above conditions, we may request formulae for how to construct these limit sets. This section will deliver the formulae for Fuchsian classical Schottky dust and all types of kissing Schottky generators, as well as special cases of some generators. In the work below, we shall choose $\psi \in [0, 2\pi]$ and $\lambda \in [0, 1]$ such that a disk D with centre c and radius r may be defined

$$D = \{c + \lambda e^{i\psi} r\}$$

6.1 Fuchsian Classical Schottky Generators

Consider a unit circle in $\hat{\mathbb{C}}$ and we shall let this be the orthogonal circle to 4 Schottky disks D_a, D_b, D_A, D_B with the distance between their centres and the origin all equal to r . Using the unit circle, we see that the orthogonality case is always true when $r = \sec(\theta)$ and the radius of a Schottky disk is $\text{Radius}(D_i) = \tan(\theta)$. We shall choose the Schottky disks of a and A to be on the imaginary axis and the Schottky disks of b and B to be on the real axis, thus the generators a, b are constructed as such:

$$a = \csc(\theta) \cdot \begin{pmatrix} 1 & i \cos(\theta) \\ -i \cos(\theta) & 1 \end{pmatrix}$$

$$b = \csc(\theta) \cdot \begin{pmatrix} 1 & \cos(\theta) \\ \cos(\theta) & 1 \end{pmatrix}$$

We observe that the trace of both maps is $2 \csc(\theta) \geq 2$, therefore all maps are guaranteed to be loxodromic. The Schottky disks for these circles are computed via the formulae for the radius and center in Proposition 7:

$$D_a = \{-i \sec(\theta) + \lambda e^{i\psi} \cdot \tan(\theta)\}, D_A = \{i \sec(\theta) + \lambda e^{i\psi} \cdot \tan(\theta)\}$$

$$D_b = \{-\sec(\theta) + \lambda e^{i\psi} \cdot \tan(\theta)\}, D_B = \{\sec(\theta) + \lambda e^{i\psi} \cdot \tan(\theta)\}$$

We must ensure that the Schottky disks associated with each map are disjoint to each other to satisfy the definition of Schottky groups in Definition 24. We shall do this by restricting the possible values of θ .

Observation 10. *For the Fuchsian classical Schottky generator formulae constructed above, the permitted values of θ such that the limit set is a Fuchsian classical Schottky dust are $\theta \bmod 2\pi \in [0, 2\pi] \setminus [0.25\pi, 0.75\pi]$.*

Proof. We shall attempt to find when the Schottky disks are to intersect. A Schottky disk's inverse disk will never intersect in this system because $\sec(\theta) \geq \tan(\theta)$, so ignore this case. Then, without loss of generality, choose to observe the disks D_a and D_b . The centres of these disks are $C(D_a) = -i \sec(\theta)$ and $C(D_b) = -\sec(\theta)$, so we can find the distance between these two points:

$$d(-\sec(\theta), -i \sec(\theta)) = \sqrt{2} \cdot \sec(\theta)$$

We also know the radii for both of these disks to be $\tan(\theta)$, therefore we can construct an inequality for when the disks do intersect:

$$2 \cdot \tan(\theta) > \sqrt{2} \cdot \sec(\theta)$$

Upon calculation, we find that $\sin(\theta) > \frac{1}{\sqrt{2}}$ and therefore $\theta \bmod 2\pi \in [0.25\pi, 0.75\pi]$. To find the permitted values of θ for Fuchsian classical Schottky dust limit sets, we must exclude these possible values of θ , thus proving the required observation. $\square$

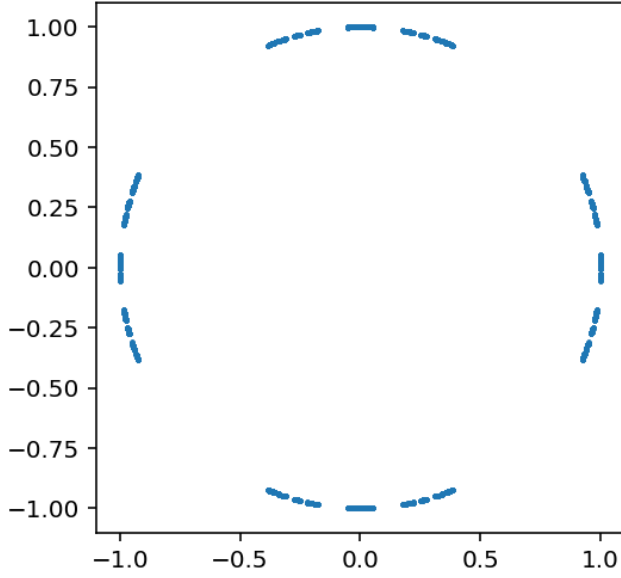


Figure 22: Fuchsian classical Schottky dust constructed with the above generators when $\theta = 0.7$.

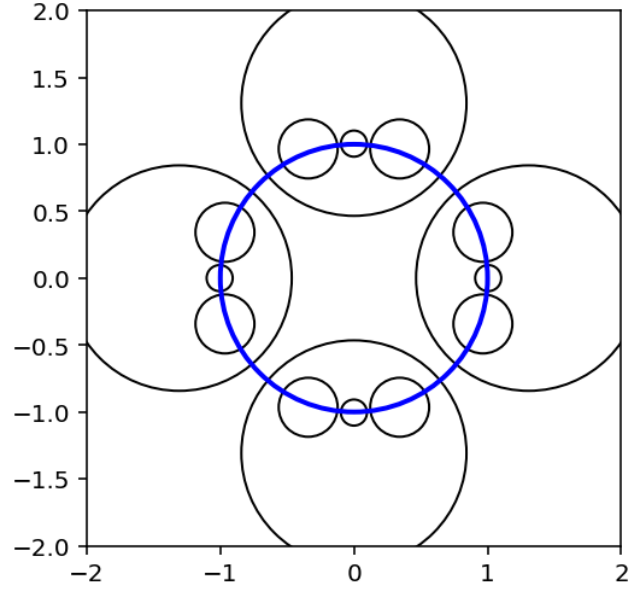


Figure 23: The first two iterations of the Schottky disks when $\theta = 0.7$, including the unit circle in blue to showcase the orthogonality of the disks.

6.2 Kissing Schottky Generators

To generate circles, we use the generators as provided in chapter 6 of Indra's Pearls. It is tasked to construct two Möbius maps such that all maps have $|\text{Tr}(M)| > 2$, have determinant 1 and combine to have parabolic commutator. It is a known result that only two variables $x, y \in \mathbb{R}$ are necessary to satisfy these properties. Define also $k = 1/xy - \sqrt{(1/xy)^2 - 1} \in \mathbb{R}$ to ensure the correct parabolic trace of $abAB$, then the generators a, b are constructed as such:

$$a = \begin{pmatrix} \sqrt{1+x^2} & ikx \\ x/ik & \sqrt{1+x^2} \end{pmatrix}$$

$$b = \begin{pmatrix} \sqrt{1+y^2} & y \\ y & \sqrt{1+y^2} \end{pmatrix}$$

It is worth noting an extra condition on y and x , where due to the equation used for k , we must have $0 < xy \leq 1$ and thus also $0 < k \leq 1$. We see that the trace a and b will always be greater than 2, therefore these generators are satisfactory to generate a kissing Schottky group.

The Schottky disks for these circles are provided below:

$$D_a = \left\{ \frac{ik\sqrt{1+x^2}}{x} + \lambda e^{i\psi} \cdot \frac{k}{x} \right\}, D_A = \left\{ \frac{-ik\sqrt{1+x^2}}{x} + \lambda e^{i\psi} \cdot \frac{k}{x} \right\}$$

$$D_b = \left\{ \frac{-\sqrt{1+y^2}}{y} + \lambda e^{i\psi} \cdot \frac{1}{y} \right\}, D_B = \left\{ \frac{\sqrt{1+y^2}}{y} + \lambda e^{i\psi} \cdot \frac{1}{y} \right\}$$

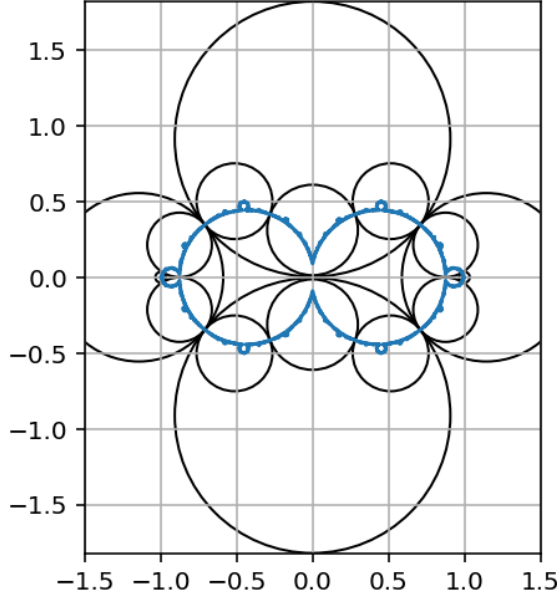


Figure 24: Generators constructed with variables $x = 1.8, y = 0.1$.

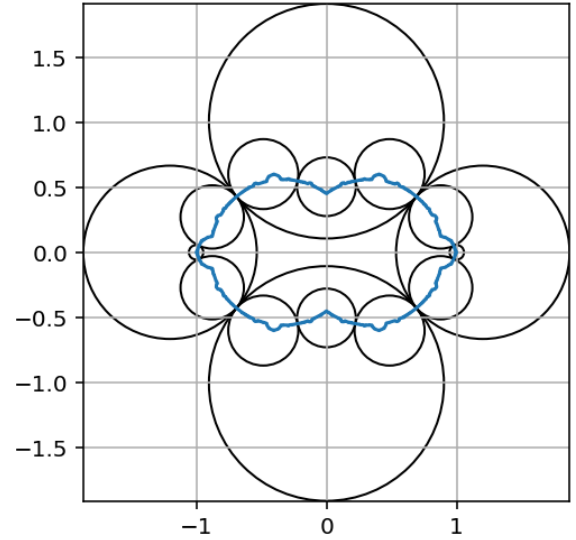


Figure 25: Generators constructed with variables $x = 1, y = 0.7$.

6.2.1 Special Cases of the Kissing Schottky Generators

Theorem 4. *Using the kissing Schottky generator formulae, a Fuchsian connected limit set is generated when $xy = 1$.*

Proof. We shall start with the assumption that $x \neq y^{-1}$ and that we have inputted two distinct values which satisfy the condition that $xy \leq 1$. Then, we can compute the traces of a and b :

$$Tr(a) = 2\sqrt{1+x^2}, Tr(b) = 2\sqrt{1+y^2}$$

Calculate the trace of ab by the formula

$$Tr(ab) = \frac{Tr(a)Tr(b) - \sqrt{(Tr(a)Tr(b))^2 - 4(Tr(a)^2 + Tr(b)^2)}}{2}$$

$$Tr(a)Tr(b) = 4\sqrt{(x^2+1)(y^2+1)}$$

$$Tr(a)^2 + Tr(b)^2 = 4(2+x^2+y^2)$$

$$Tr(ab) = \frac{1}{2} \left[\sqrt{(x^2+1)(y^2+1)} - \sqrt{(x^2+1)(y^2+1) - 16(2+x^2+y^2)} \right]$$

From this, consider the large square root on the right and determine the rough boundaries of when the root is imaginary:

$$\begin{aligned} y^2 x^2 - 15(y^2 + x^2) - 31 &< 0 \\ y^2(x^2 - 15) &< 15x^2 + 31 \\ y^2 &< \frac{15x^2 + 31}{x^2 - 15} \end{aligned}$$

Due to the symmetry of the equation, such is the case also for x :

$$x^2 < \frac{15y^2 + 31}{y^2 - 15}$$

From this, we may see that for the above fractions, there are clear asymptotes at $x, y = \sqrt{15}$, thus the square root is negative for values $x, y < \sqrt{15}$. If we choose x, y such that they are not within this boundary, then $xy > 1$ and the value of k is not real.

As a special case, however, consider when x, y are related such that $y = x^{-1}$. In this case, $xy = 1$ and $k = 1$. Following with calculating the traces of maps like earlier:

$$\begin{aligned} Tr(a) &= 2\sqrt{1+x^2} \in \mathbb{R}, Tr(b) = 2\sqrt{1+x^{-2}} \in \mathbb{R} \\ Tr(a)Tr(b) &= 4\sqrt{(1+x^2)(1+x^{-2})} = 4\sqrt{(x+x^{-1})^2} = 4(x+x^{-1}) \\ Tr(a)^2 + Tr(b)^2 &= 4(1+x^2) + 4(1+x^{-2}) = 4(x+x^{-1})^2 \end{aligned}$$

$$Tr(ab) = \frac{1}{2} \left[4(x+x^{-1}) - \sqrt{16(x+x^{-1})^2 - 16(x+x^{-1})^2} \right] = 2(x+x^{-1}) \in \mathbb{R}$$

Therefore, we have shown that for all values of x and $y = x^{-1}$, there is no case where $Tr(ab) \notin \mathbb{R}$, resulting in the traces of a, b, ab all being real-valued for all x . We see that all maps have trace greater than 2 and are therefore loxodromic because of the observation

$$\min\{x + x^{-1} : x \in \mathbb{R}\} = 2 > 1.$$

These cover all of the conditions required in Definition 30 for when a Schottky group's associated limit set is Fuchsian. Since the case of the parabolic commutator is already accounted for in the kissing Schottky generator formulae, we have successfully proven that the limit set is a Fuchsian connected limit set. $\square$

It follows that choosing this formulae for generating Möbius maps and applying the extra condition $xy \neq 1$ assures that the limit set is quasi-Fuchsian containing infinitely small details and has dimension $\dim_H(\Lambda) > 1$.

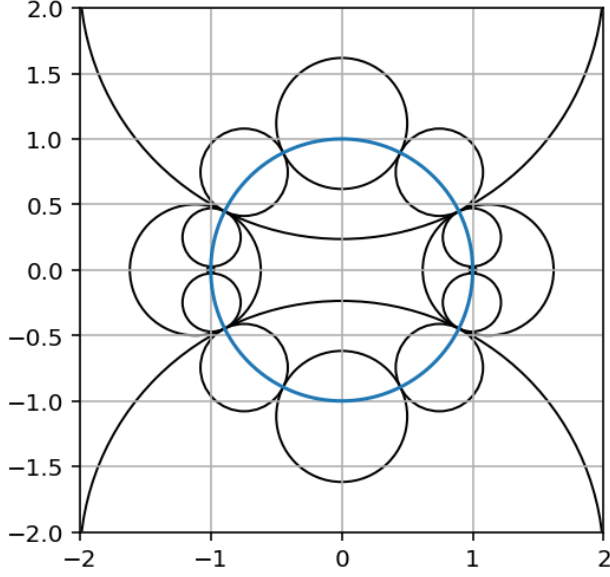


Figure 26: Generators constructed with variables $x = 0.5, y = 2.0$. The set is Fuchsian with $\dim_H(\Lambda) = 1$ because $xy = 2 \cdot 0.5 = 1$.

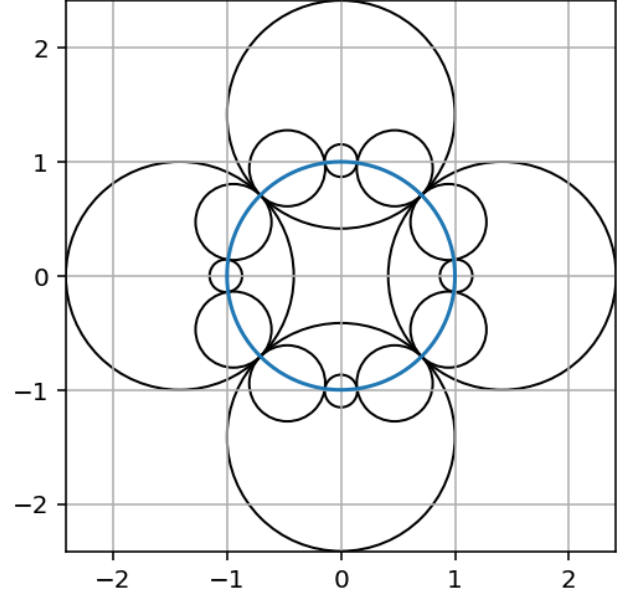


Figure 27: Generators constructed with variables $x = y = 1$. Since $xy = 1$, the limit set is Fuchsian and of dimension $\dim_H(\Lambda) = 1$.

6.3 Trace-Input Kissing Schottky Generators

The limitation of the kissing Schottky generators in Section 6.2 is that it is not possible for the trace of a map to be equal to 2. As a result, the limit sets associated with a Schottky group such that either a or b are parabolic are impossible to produce. Furthermore, we are unable to input any non-real values into the previous formulae, restricting our possible dynamics to those which are more simple. Instead, we may find that there exists a formulae for generators depending on solely the traces of the two maps a and b . To do so, we simply require the inputs $t_a = \text{Tr}(a), t_b = \text{Tr}(b)$ and we may then solve for $t_{ab} = \text{Tr}(ab)$, using the same formula as in Theorem 4, and the fixed point of the commutator $z_0 : abAB(z_0) = z_0$. The generators are then constructed as below:

$$a = \begin{pmatrix} \frac{t_a}{2} & \frac{t_a t_{ab} - 2t_b + 4i}{(2t_{ab} + 4)z_0} \\ \frac{(t_a t_{ab} - 2t_b - 4i)z_0}{2t_{ab} - 4} & \frac{t_a}{2} \end{pmatrix}$$

$$b = \begin{pmatrix} \frac{t_b - 2i}{2} & \frac{t_b}{2} \\ \frac{t_b}{2} & \frac{t_b + 2i}{2} \end{pmatrix}$$

Due to the complexity in the formulae of generators above, the equations for the Schottky disks are omitted. Generally, it is not suitable to work with the disks associated with these maps and it is recommended to simply work with limit points and the limit set.

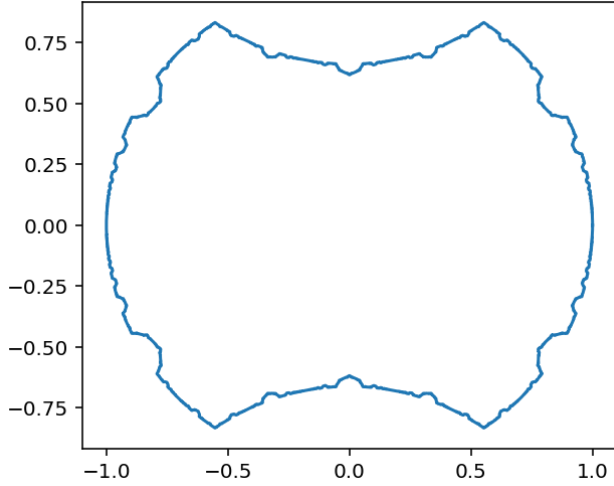


Figure 28: Generators constructed with traces $\text{Tr}(a) = 3$, $\text{Tr}(b) = 2.4$.

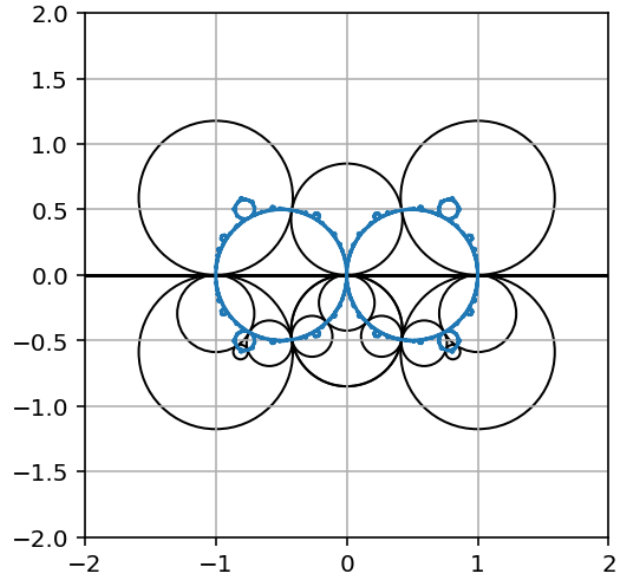


Figure 29: Generators constructed with traces $\text{Tr}(a) = 2$, $\text{Tr}(b) = 3.4$.

7 Concluding Notes

In this BSc project, we have studied the topics at the intersection of real analysis, measure theory, fractal dynamics, and more. After developing the required prerequisite knowledge through exercise sheets, we first introduce rigorous foundation in measure theory and measurable sets. Afterwards, our focus has been shifted towards the study of Möbius maps on the extended complex plane, mainly their classification by their dynamical properties. We then used Möbius maps to introduce Schottky groups generated by an even number of these maps, as well as the limit sets associated with a Schottky group. Finally, we explored the different types of limit sets and the formulae for the generating Möbius maps with a limit set of a given type.

In this project, focus has been given to understanding the notion of dimension and bounds on dimension of a wide range of sets. We start with textbook examples for box dimensions and Hausdorff dimensions of the Cantor set, Sierpinski triangle and Bedford-McMullen carpets which we have reworked. Subsequently, we give an overview of the existing results related to the bounds of Hausdorff dimension for limit sets associated with Schottky groups.

In summary, we have developed solid foundations for studying the Hausdorff dimension of quasi-Fuchsian and parabolic limit sets, which is a case of special interest as they cannot be tackled by the present methods.

References

- [1] Patrick Billingsley, **Probability and Measure**, *John Wiley and Sons*, 1995
- [2] Rufus Bowen, **Hausdorff Dimension of Quasi-Circles**, *Publications Mathématiques de l'IHÉS*, 1979
- [3] Jean-Francois Colonna, Available at <https://www.lactamme.polytechnique.fr/Mosaic/images/SIER.51.012345.D/display.html>
- [4] Jean-Francois Colonna, Available at <https://www.imaginary.org/gallery/jean-francois-colonna-deterministic-fractal-geometry>
- [5] Gerald Edgar, **Classics on Fractals**, *Westview Press*, 1993
- [6] Kenneth Falconer, **Fractal Geometry: Mathematical Foundations and Applications (2nd ed.)**, *John Wiley and Sons*, 2003
- [7] Gerald Folland, **Real Analysis: Modern Techniques and Their Applications**, *John Wiley and Sons*, 2013
- [8] Paul Halmos, **Measure Theory**, *Springer New York*, 1950
- [9] John Hutchinson, **Fractals and Self-Similarity**, *Indiana University Mathematics Journal* vol. 30, no. 5, 1981, p.713-747
- [10] István Kolossváry, **An Upper Bound for the Intermediate Dimensions of Bedford-McMullen Carpets**, *Journal of Fractal Geometry, Mathematics of Fractals and Related Topics* vol. 9, 2022, p.151-169
- [11] Vladimir Kisil, **Geometry of Möbius Transformations: Elliptic, Parabolic and Hyperbolic Actions of $SL(2, \mathbb{R})$** , *Imperial College Press*, 2012
- [12] Benoit Mandelbrot, **Les Objets Fractals: Forme, Hasard et Dimension**, *Flammarion (Paris)*, 1975
- [13] Benoit Mandelbrot, **Fractals: Form, Chance, and Dimension**, *W. H. Freeman and Company (San Francisco)*, 1977
- [14] Benoit Mandelbrot, **The Fractal Geometry of Nature**, *W. H. Freeman and Company (New York)*, 1982
- [15] Martin McBride, **Cantor Set**, 2023, Available at <https://graphicmaths.com/fractals/fractals/cantor-set/>
- [16] David Mumford, Caroline Series and David Wright, **Indra's Pearls: The Vision of Felix Klein**, *Cambridge University Press*, 2002
- [17] Tristan Needham, **Visual Complex Analysis**, *Oxford University Press*, 1997
- [18] Hemant Kumar Pathak, **Complex Analysis and Applications**, *Springer Singapore*, 2019

- [19] Heinz-Otto Peitgen, **The Beauty of Fractals: Images of Complex Dynamical Systems**, *Springer-Verlag (Heidelberg/New York)*, 1986
- [20] Larry Riddle, **Classical Iterated Function Systems**, Available at <https://larryriddle.agnesscott.org/ifs/siertri/siertri.htm>
- [21] Bernhard Riemann, **Theorie der Abel'schen Functionen (Theory of Abelian Functions)**, 1857
- [22] Walter Rudin, **Principles of Mathematical Analysis**, *McGraw-Hill*, 1976
- [23] Stanislav Smirnov, **Dimension of Quasicircles**, *Acta Mathematica vol. 205*, 2010, p.189-197
- [24] Polina Vytnova and Caroline Wormell, **Hausdorff Dimension of the Apollonian Gasket**, *Inventiones Mathematicae vol. 239, no. 3*, 2025, p.1003-1068